

一种面向情感语言模型游戏智能体的基于评价的情绪链架构

预印本

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摘要

开发逼真、自然且具有交互性的数字人工代理是一个日益受到关注的领域。理论上的不确定性以及技术上的障碍给该领域带来了巨大挑战, 尤其是在开发能够有效模拟人类情感的智能体方面。大语言模型或许能够通过挖掘情境评估中的通用模式来应对这些问题。本研究通过三项实证实验, 测试了大语言模型解决情商任务和模拟情感的能力。它基于心理评估研究, 提出并评估了一种用于视频游戏中情感模拟的新颖情感链架构。实验结果表明, 在用户体验和内容分析指标方面, 该架构优于标准的大语言模型架构。因此, 本研究为如何构建和测试基于语言模型中表征的认知过程的情感智能体提供了早期证据。

关键词 L 大语言模型 · 情感计算 · AI智能体 · 情感 · 模拟 · 视频游戏

1 引言

在以用户为中心的软件和视频游戏中, 情感人工智能体因其在娱乐和训练场景中提供个性化、自然且引人入胜的体验的潜力而受到长期研究 [46, 25, 48]。情感人工智能体主要通过其在特定情境下模拟适当情感响应的能力来定义 [6, 25]。因此, 它们被认为有助于提升各领域中的用户交互体验 [6], , 甚至可能带来健康益处 [36]。

然而, 构建能够成功建模并表达情感的系统是一项艰巨的任务 [46], , 尤其是在心理学研究中, 情感和情感本身也是模糊的概念 [52]。情感过程非常复杂, 涉及多个组成部分 (如认知、行为、生理和感受 [53]), , 这些部分尚未被完全理解, 甚至在基本层面上存在广泛争论 [28], , 这使得计算表征变得非常困难 [25]。随着现代技术的发展, 这类基于机器学习 [57], 的情感模拟或许可以通过数据驱动技术实现。

例如, 大语言模型已展现出模拟一系列认知能力的潜力[7], 甚至能向他人推断多种心理和情感状态 [30]。由于大语言模型是在包含人类心智能力表征的大规模文本数据上训练的, 因此它们已被观察到能在多种任务上表现出类人水平的表现 [7, 23]。既然情感是人类感知现实的重要组成部分, 从而

AN APPRAISAL-BASED CHAIN-OF-EMOTION ARCHITECTURE FOR AFFECTIVE LANGUAGE MODEL GAME AGENTS

PREPRINT

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ABSTRACT

The development of believable, natural, and interactive digital artificial agents is a field of growing interest. Theoretical uncertainties and technical barriers present considerable challenges to the field, particularly with regards to developing agents that effectively simulate human emotions. Large language models (LLMs) might address these issues by tapping common patterns in situational appraisal. In three empirical experiments, this study tests the capabilities of LLMs to solve emotional intelligence tasks and to simulate emotions. It presents and evaluates a new chain-of-emotion architecture for emotion simulation within video games, based on psychological appraisal research. Results show that it outperforms standard LLM architectures on a range of user experience and content analysis metrics. This study therefore provides early evidence of how to construct and test affective agents based on cognitive processes represented in language models.

Keywords Large Language Models · Affective Computing · AI Agents · Affect · Simulation · Video Games

1 Introduction

In user-centered software and video games, affective artificial agents have long been researched for their potential to provide personalized, natural, and engaging experiences for both entertainment and training purposes [46, 25, 48]. Affective artificial agents are mainly defined through their ability to simulate appropriate emotional responses given certain situations [6, 25]. Consequently, they are believed to contribute to enriched user interactions on various domains [6], and even potential health benefits [36].

However, building systems that successfully model, and express emotions is a difficult task [46], especially since affect and emotion are fuzzy concepts, even in psychology research [52]. Affective processes are very complex and involve multiple components (such as cognitive, behavioural, physiological, and feeling [53]) that are not fully understood and often debated on a fundamental level [28], which makes computational representations very difficult [25]. With the development of modern technology, such machine learning [57], emotion simulation might be achievable through data-driven techniques.

For example, large language models (LLMs) have demonstrated a potential to simulate a range of cognitive abilities [7] and even imputing a range of mental and affective states to others [30]. Because LLMs are trained on large text bodies that hold representations of human mental abilities, they have been observed to exhibit human-like performance on a variety of tasks [7, 23]. Since emotions are an important part of how humans perceive reality and therefore

构建语言 [5]，并且深受认知过程的影响 [39]，包括语言标注 [34, 10]，基于语言的情感表征或许也能让深度学习模型更好地模拟情感反应。然而，迄今为止，大语言模型在解决情感智能体领域中现存问题的潜力尚未得到充分理解。

因此，本研究考察了大语言模型模拟情感的潜力，以及这种潜力如何受到底层实现架构的影响。我们利用情感研究的最新发现，提出了一种基于认知评价的语言模型情感生成方法，并与其他策略进行对比测试，评估其生成恰当情境情感的能力。随后，我们利用这些结果，在一个新开发的对话式视频游戏中实现了情感智能体。因此，本研究旨在利用语言模型推进人工智能体的情感模拟进程。

2 相关工作

2.1 情感智能体

在情感计算领域，研究人员和开发者致力于创建能够智能响应玩家情感变化的情感系统 [46]。将情感计算技术应用于视频游戏的一些优势包括：为不同玩家提供更一致、更易获得的玩家体验 [21]，个性化健康与训练应用 [3]，以及旨在强化目标体验的、全新且精心设计的游戏机制 [24, 64]。情感智能体在视频游戏中的应用研究尤其侧重于这最后一个目标。2011年，Hudlicka讨论了情感（或更准确地说，情绪）游戏智能体的潜在系统设计要素[25]。根据该作者的观点，情感智能体可以被视为对情感理论模型计算实现的一种表达，这些模型具备用于情感生成的评价功能。例如，人工智能体可以对特定事件进行计算，以评估其与智能体的关联性，从而得出可能的情感反应 [25]。“计算友好型”评价实现通常建立在诸如OCC模型 [42]（例如 GAMYGDALA [47]）等模型之上。在考虑了特定的固定因素（例如智能体的期望 [9]）后，此类模型已被用于基于决策树模拟评价过程。

此类人工智能体的主要目标是呈现出自然、类人行为与可信度，作为更丰富、更具沉浸感的游戏世界的一部分 [24, 25]。因此，智能体的任务有别于其他情感游戏机制——后者主要试图让游戏世界适应玩家情感 [65, 16]。基于情感信息的程序化生成内容（PCG）已被证明能在视频游戏中成功提升乐趣，并提供个性化、沉浸式的体验（例如可参考 Shaker 等人的工作 [56, 55]）。这通常通过微调某些与目标玩家情绪相关的机制来实现，以提高该情绪出现的概率 [65]。然而，情感智能体并不需要根据玩家情绪调整行为，而是需要拥有自身的情感表征，进而衍生出可信的行为（或其他情感组成部分的自然表征，例如模拟感受或模拟生理 [25]）。

因此，设计和开发情感游戏智能体的核心问题在于创建良好的情绪状态计算模拟。人类情绪是复杂的心理生理状态，通过行为、生理、认知和感受等组成部分得以表达 [52]。此外，虽然已有大量工作对情绪进行实证研究，但许多核心理论分歧依然存在 [52]，包括维度论 [49]、离散论 [27]、建构论 [5]，和认知论 [53] 等不同视角之间的争论。

一个完全成熟的情感智能体要求首先解决自情感计算诞生以来一直存在的所有基本心理学空白 [46]，然后将它们整合到可运行的计算系统中 [25]。这意味着，为人工智能体构建一个基于心理学、功能完备且准确的情感模拟，目前尚不可行，并且在几乎所有游戏设计案例中都不切实际。然而，我们仍然可以构建具备人类情绪诱发关键特征的情感智能体，从而实现对人类情感的相对成功模拟。其中一个候选特征是评估。情绪诱发依赖于情境因素和个体因素，通过评估进行处理 [39, 54]。情绪评估的概念是指，情绪是由对触发事件的主观评价所引起的，这些评价涉及这些事件对个人生活或兴趣的重要性 [33]。有证据表明，评估在情绪诱发中起着核心作用，并因此影响所有其他情感组成部分 [54]。

任何给定的外部事件（例如情境）或内部事件（例如想法）都可能在多个变量上被评估，这些变量有助于情感的形成。这些变量可能包括目标相关性、确定性、应对潜力或主体性 [39]。因此，评估代表了一个适应个体差异 [22] 和当前上下文 [38]的灵活过程。证据还表明，语言可以在情绪评估中发挥关键作用，既通过提供构建评估所需的关键上下文信息，也通过为被评估的状态提供概念标签 [10]。

construct language [5] and are heavily influenced by cognitive processes [39] including linguistic labelling [34, 10], language-based emotion representations might too enable deep learning models to better simulate affective responses. As of yet, the potential for LLMs to solve some of the issues present in the field of affective agents is however not well understood.

This study therefore examines the potential of LLMs to simulate emotions and how this potential might be influenced by the underlying implementation architecture. Using contemporary findings of emotion research, we propose a cognitive appraisal-based approach for language model affect generation and test it against other strategies in the ability to generate appropriate situational emotions. We then use those results to implement affective agents within a newly developed conversational video game. This study therefore represents an effort to progress affect simulation for artificial agents using language models.

2 Related Work

2.1 Affective Agents

In affective computing, researchers and developers are interested in creating affective systems that intelligently respond to changes in users’ emotions [46]. Some of the benefits associated with affective computing techniques applied to video games include more consistent, accessible player experience for a range of different players [21], personalized health and training applications [3], as well as new and purposefully designed gameplay mechanisms aimed at reinforcing target experiences [24, 64]. The use of affective agents in video games has been researched with special regard to this last aim. In 2011, Hudlicka discussed potential system design elements for affective (or more precisely emotional) game agents [25]. According to the author, affective agents can be seen as computational representations of operationalizations made from emotion-theoretical models with appraisal functionality for emotion generation. For example, artificial agents could implement computational calculations of certain events to assess the relevance to the agent and consequently probable emotional reaction [25]. "Computation-friendly" appraisal implementations have often been built on models such as the OCC model [42] (see for example GAMYGDALA [47]). Taken specific fixed aspects (such as expectations of the agent [9]) into account, such models have been used to simulate appraisal based on decision trees.

The main aim for such artificial agents is seen to be natural, human-like behaviour and believability as a part of a more fleshed-out and engaging game world [24, 25]. The tasks of agents therefore differ from other affective game mechanism that mostly try to adapt the game world to player affect [65, 16]. Procedurally generated content (PCG) based on affective information has been shown in video games to successfully increase enjoyment and offer personalized, immersive experience (see for example the work of Shaker et al. [56, 55]). This is often done by fine-tuning certain mechanics shown to be associated with a target player emotion to increase the probability for that emotion [65]. Affective agents however do not need to adapt behaviours to player emotions, but rather need their own emotion representations that could then lead to believable behaviours (or other natural representations of emotion components, such as simulated feeling or simulated physiology [25]).

The central issues of designing and developing affective game agents lies therefore in creating good computational simulations of emotional states. Human emotions are complex psycho-physiological states that are expressed within behavioural, physiological, cognitive, and feeling components [52].Moreover, while much work has been done to empirically investigate emotions, many core theoretical disagreements remain [52], including debates between dimensional [49], discrete [27], constructivist [5], and cognitive [53] perspectives.

A fully developed affective agent would make it necessary to first solve all fundamental psychological gaps that had been present since the beginning of affective computing [46] and then integrate them into working, computational systems [25]. This means that building a psychology-based, fully functional and accurate emotion simulation for an artificial agent is currently not possible and would be in almost all game design cases impractical. However, we may still be able to build affective agents that possess key features of emotion elicitation in humans and, as a consequence, allow for relatively successful simulation of human emotions. Once candidate feature is appraisal. Emotion elicitation is dependent on contextual and individual factors, processed through appraisal [39, 54]. The notion of emotion appraisal is that emotions are caused by subjective evaluations of triggering events in regards to their significance to one’s personal life or interests [33]. Evidence suggests that appraisal holds a central role in emotion elicitation and as a consequence acts on all other emotion components [54].

Any given external (e.g. situations) or internal (e.g. thoughts) event may be appraised on multiple variables that contribute to emotion forming. Such variables might include goal relevance, certainty, coping potential, or agency [39]. Appraisal therefore represents a flexible process that adapts to individual differences [22] and the current context [38]. Evidence also suggests that language can play a key role in emotional appraisal, both by providing key contextual information from which to construct the appraisal and by providing conceptual labels for the appraised states [10].

考虑到这些因素，语言模型可能提供了一种模拟评估过程的手段，因为它们能够基于文本训练数据生成高层次、意义驱动的文本输出，这些数据可能潜在地包含人类心理过程的隐式表征 [7]。

2.2 语言模型方法

在过去几年中，自然语言处理（NLP）迅速发展，以至于单一的任务无关语言模型能在多项任务中表现良好 [11]，包括模拟类人行为 [23]。这背后的基础是海量的训练数据，这些数据通过语言代表了广泛的人类行为[11, 8]。在游戏方面，OpenAI的生成式预训练变换器（GPT-2及其后续版本）等模型已在交互式故事的程序化生成 [20]、基于文本的冒险对话以及动作候选 [66, 12]方面取得了初步成功。在Park 等人 [44], 近期的一项模拟研究中，语言模型被嵌入人工智能体架构中，用于填充一个类似于《模拟人生》[4]的沙盒世界。该架构包括根据与当前情境的相关性，从记忆系统中存储和检索信息，并利用这些信息生成反思（即对情境的高层解读）、计划（即潜在的未来行动）以及即时动作。多个智能体在类游戏世界中进行了模拟，作者指出，就类人模拟而言，所涌现出的交互自然且可信。

尽管该领域的研究仍处于早期阶段，但语言模型的使用解决了先前方法的一些问题。最值得注意的是，主要任务不再试图构建人类行为的计算表示，而是从语言模型中针对给定情境检索可信的人类行为，并将实验结果集成到游戏智能体中。根据游戏目标，这涉及：（1）将当前情境与预期结果转换成语言；（2）使用大型语言模型生成内容；以及（3）将输出转换回来，以便在游戏系统中实现。例如，Ciolino 等人 [13] 使用微调的 GPT-2 模型生成围棋走法，方法是将当前棋盘状态转换为文本，再将语言输出转换回行动建议。这一过程对于纯文本任务（例如对话生成）自然更为容易，因为文本已经是预期输出，且预期输出相对容易用语言描述 [66, 44]。

尽管如此，即使是纯文本的生成任务也可能给语言模型带来一些潜在障碍。最明显的障碍来自底层的训练数据。没有任何一个语言模型能够完整地代表人类行为，它只能局限于训练数据及其偏差 [60] 以及模型规范 [11]。此外，语言模型的性能不仅取决于输出，也取决于输入 [61]。例如，思维链提示是由谷歌研究团队的 Wei 等人 [62] 提出的一个概念，它涉及将思维链融入少样本提示中，以提升语言模型的推理能力。同样，如 Park 等人 [44] 所述，在数字世界中模拟可信行为包含多个重要步骤（包括存储和检索记忆、在观察之外进行反思等），这些步骤最终协同作用，以提高生成符合预期的自然行为的概率。

在设计情感智能体（即模拟情感的智能体）时，我们首先需要问的是：情感在训练数据中的代表性如何，以及语言模型能否检索到它？在最近一篇讨论不同 GPT 版本在心智理论任务上表现的文章中，Kosinski [30] 发现，新的语言模型在将不可观察的心理和情感状态归因于他人方面表现非常出色。这些发现（尤其是结合表明在认知任务上表现良好的发现 [7, 58]）表明，高级心理机制仅凭语言就能得以体现，因此可以通过构建良好的语言模型进行模拟。沿着这个思路，我们能否有效且高效地利用语言模型实现准确且自然的情感模拟？如果我们能假设情感在语言模型中有所体现，那么情绪诱发机制（如评估）也可能同样存在。并且，鉴于语言模型可以通过上下文学习 [11, 61]，（例如通过思维链提示 [62], ）得到改进，情感生成或许能通过允许情感上下文学习的架构得到促进。因此，本研究通过测试不同实现架构的情感生成能力，来探讨语言模型模拟情感游戏智能体的潜力，其中包含一种新开发的基于评价的架构，旨在促进自然的情感链。

3 评估提示策略用于情感模拟

本项整体研究的目的是为对话游戏（即基于语言的用户输入和基于语言的智能体响应的游戏）创建一种有效的情感智能体架构。然而，由于语言模型已被成功用于生成智能体动作空间 [66, 44]，这一过程也可应用于非玩家视频游戏角色中其他类似人类情感的模拟。

With all this in mind, language models might provide one mean of simulating the appraisal process as they are able to generate high-level meaning-driven text outputs based on text training data that potentially holds implicit representations of human psychological processes [7].

2.2 Language Model Approach

In the last few years, Natural Language Processing (NLP) has been rapidly progressing to the point that singe task-agnostic language models perform well in a range of tasks [11], including the simulation of human-like behaviour [23]. The basis for this is the large amount of training data representing a wide range of human behaviour through language [11, 8]. In relation to games, models such as OpenAI’s Generative Pre-trained Transformer (GPT-2 and its successors) have shown early successes in the procedural generation of interactive stories [20], text-based adventure dialog as well as action candidates [66, 12]. In a recent simulation study by Park et al. [44], language models were implemented in artificial agent architectures to populate a sandbox world reminiscent of The Sims [4]. The architecture includes storing and retrieving information from a memory system and based on relevancy for the current situation and then uses the information to generate reflections (i.e. high-level interpretations of a situation), plans (i.e. potential future actions), and immediate actions. Multiple agents were simulated in a game-like world and the authors suggest that emerging interactions were natural and believable in terms of human-like simulation.

While work in this area is still in an early stage, the use of language models addresses some concerns with prior approaches. Most notably, instead of trying to build computational representations of human behaviour, the main task involves trying to retrieve believable human behaviour given a situation from a language model and implementing the results within a game agent. Depending on the game aim, this involves (1) translating the current situation with regards to the expected outcome into language; (2) generating content using a large language model; and (3) translating the output back in order to implement it in a game system. For example, Ciolino et al. [13] used a fine-tuned GPT-2 model to generate Go moves by translating the current board state to text and the language output back to action suggestions. Such a process is naturally easier for purely text-based tasks, such as dialog generation, where text is already the expected output and the expected output can comparatively easily be described in language [66, 44].

Still, even purely text-based generative tasks can pose some potential barriers for language models. The most obvious barrier comes from the underlying training data. No language model can represent human behaviour in its entirety, but is limited to the training data and its biases [60] as well as model specifications [11]. Additionally, performance of language model is not only dependent on the output, but also on the input [61]. For example, chain-of-thought prompting is a concept introduced by Wei et al. from the Google Research team [62] and relates to the integration of chain-of-thought in few-shot prompts to improve reasoning capabilities of language models. Similarly, as Park et al. [44] describe, simulating believable behaviour in a digital world includes various important steps (including storing and retrieving memory, reflecting in addition to observing, etc.) that ultimately work together to improve the probability to generate expected and natural behaviour.

When it comes to designing affective agents (i.e. agents that simulate emotions), the first questions that we have to ask is how well is affect represented in the training data and is a language model capable of retrieving it? In a recent paper discussing performance of different GPT iterations on theory of mind tasks, Kosinski [30] found that new language models perform very well when it comes to imputing unobservable mental and affective states to others. Such findings (especially combined with findings suggesting good performance on cognitive tasks [7, 58]) suggest that high-level psychological mechanisms are represented in language alone and could therefore be simulated with a well-constructed language model. Along these linses, can we effectively and efficiently achieve accurate and natural affect-simulation using language models? If we can assume that emotions are represented in language models, mechanisms for emotion elicitation (such as appraisal) might also be represented. And given that language models can be improved through in-context learning [11, 61], for example by chain-of-thought prompting [62], affect generation might be facilitated by architectures that allow for affective in-context learning. This study therefore discusses the potential of language models to simulate affective game agents by testing affect generation capabilities of different implementation architectures, including a newly developed appraisal-based architecture to facilitate natural chain-of-emotion.

3 Appraisal-Prompting Strategy for Emotion Simulation

The aim of this overall study is to create an effective affective agent architecture for a conversational game (i.e. a game with language-based user input and language-based agent response). However, since language models have also been successfully used to generate agent action spaces [66, 44], this process could also be applicable to other simulations of human-like affect in non-playable video game characters.

该方法的基础植根于传统的程序化游戏内容研究，尤其是整合了情感适应（例如参见[65, 55]）。用户输入（在本例中为文本输入）通常会被解析到游戏逻辑中，以有意义的方式调整内容，例如更好地激发目标体验 [64]。这可能意味着游戏智能体会对某些玩家输入甚至它们在游戏世界 [44] 中的自身交互做出反应。为了利用语言模型的功能，交互需要被转化为语言，并且根据所使用的具体语言模型，语言应遵循特定模式以产生最佳结果，这通常被称为提示工程 [63]。一个可被视为与模拟游戏智能体固有相关的模式是角色模式，它指示大语言模型扮演某个特定的角色。结合游戏模式中为角色任务提供抽象游戏上下文 [63]，的方面，使用此类模式合成的交互最基本形式仅包含用于生成响应的玩家提供的文本。由于情感在语言模型 [30]，中有所体现，仅这一非常基础的步骤就可以构建一个初级的情感智能体。

然而，静态提示模式在创建可信的游戏智能体方面存在局限性。最值得注意的是，它们没有将记忆作为人类互动的基本基础。集成语言模型（如ChatGPT）的应用通过记录输入-响应历史来部分解决这一问题，该历史会影响后续内容的生成，从而可以创建更自然的对话流程并提升未来生成的性能 [11, 62]。在其最基本的形式中，记忆可以将之前的游戏内观察（例如玩家与智能体对话的过程）整合到后续提示中。换句话说，它扩展了提示模式，以包含记忆中的游戏内观察，从而促进上下文学习 [61]。然而，这有两个主要限制：首先，提示在可能长度和复杂性方面是有限的，这意味着无法在每次生成的提示中都包含游戏智能体的完整特征描述。Park 等人 [44] 通过设计一个记忆系统解决了这个问题，该系统在为大语言模型构建提示之前，首先从数据库中检索相关条目。另一个适用于简单模拟、资源密集度较低的解决方案可以是：仅在认为新信息对未来生成相关时才进行存储，并让记忆能够代表所有将用于提示的相关信息。第二个限制是缺乏更深层次的理解。仅追踪观察结果使得语言模型很难对场景进行推理或做出概括性解读 [7]。这个问题可以通过引入其他类型的记忆来解决，例如反思和计划 [44]。对于情感智能体而言，除了外部观察之外，更合适的追踪信息是其内部情绪进程——即情感链。

总而言之，基于语言的情感游戏智能体需要某种能够存储观察结果和情感的记忆系统。这个记忆系统是未来提示的基础。对于简单的游戏，例如本研究开发的短对话游戏，只有相关信息会被存储在记忆中，这取代了检索系统，因为游戏智能体在记忆存储的时间点可考虑的预期动作池有限。然而，模拟智能体行为的更复杂游戏应当考虑使用记忆检索系统 [44]。

为了存储情感并因此创建情感链，该架构需要一个能将观察结果转化为情绪反应的系统。由于情绪诱发高度依赖于对当前情境和个体差异进行评估的评估过程 [39, 54]，，因此该系统可以利用评价提示，即利用上下文信息和角色描述让智能体评估当前情境，以生成当前情感。如图1所示，初始上下文信息和角色信息可以由游戏设计师提供，并存储在情感智能体的记忆系统中。然后，评价系统会扩展存储的回忆，为每个观察到的行为包含当前情感，从而创建情感链。反过来，这可以用于生成智能体的行为（具体到对话游戏，即智能体的对话）。

在开发情感智能体时需要考虑的一个方面是评估标准。与大多数认知能力不同[7]，，目前成功的情绪模拟仅缺乏少数标准化基准。在心理学中，衡量评估与情感表达能力的一种方法被称为情商（EI） [51]。情商被认为是一种能力，受认知影响但并非由认知主导 [41]，，因此常被用于评估儿童和成年人在不同情境下的情感能力 [43]。然而，情商作为一个构念的定义较为模糊，许多测量方法被批评为是在情商这个总括术语下测量了可能不同的能力 [15]。因此，MacCann 和 Roberts [37] 开发了针对更精确界定维度的新测量方法，其中包括情境情绪理解测试（STEU），该测试关乎评估给定情境中最恰当情感的能力。

虽然可以说情绪理解是情感智能体必须具备的核心能力，但在情感游戏的背景下，用户体验变得更为重要。例如，游戏智能体的一个核心目标是展现可信且类人的行为 [9] 或人格与社会存在感 [50]。因此，理解和生成更精确情绪的能力只是评估视频游戏中情感模拟成功与否的一个方面，还需要研究更多以用户为中心的方法。因此，所提出的架构将在多个领域进行测试，包括情绪理解、智能体可信度，以及用户感知的情商、温暖和能力。

The basis of the approach is rooted in traditional PGC research, especially integrating affect-adaptation (see for example [65, 55]). User input (which in this case is text input) typically gets parsed into the game logic to adapt the content in a meaningful way, for example to better elicit a target experience [64]. This could mean that game agents react to certain player inputs or even their own interactions in the game world [44]. To make use of language model functionality, interactions need to be translated into language and depending on the specific language model in use, the language should follow specific patterns to yield the best result, which is generally known as prompt engineering [63]. One pattern that can be considered inherently relevant to simulating game agents are persona patterns, which instruct a LLM to assume a certain characterized role. Combined with aspects of game play pattern that provide the abstract game context for the persona tasks [63], the most basic form of an interaction synthesized with such patterns only includes player provided text used to generate responses. Because emotions are represented in language models [30], this very basic step alone could make a rudimentary affective agent.

However, static prompt patterns have limitations for creating believable game agents. Most notably, they do not incorporate memory as a basic foundation of human interaction. Applications that integrate language models, such as ChatGPT, partially address this by logging an input-response history that influences progressive content generations, which can create more natural conversation flows and improve performance of future generations [11, 62]. In its most basic form, memory could integrate the preceding in-game observations (such as the course of a player-agent dialog) into following prompts. In other words, it expands the prompt pattern to include memorized in-game observations to facilitate in-context learning [61]. This has however two major constraints: First, prompts are limited in terms of possible length and complexity, meaning that a full characterisation of a game agent cannot be included in a prompt for every given generation. Park et al. [44] addressed this problem by designing a memory system that first retrieves relevant entries from a data base before constructing prompts for the language model. Another less resource-intensive solution for simpler simulations could be to only store new information if it is considered relevant for future generations and have memory be representative of all relevant information to be used in prompts. A second limitation is a lack of deeper understanding. Tracking only observations makes it hard for a language model to draw inferences or make generalized interpretations of a scenario [7]. This problem could be addressed by introducing other types of memories, such as reflections and plans [44]. For affective agents, the more appropriate information to track in addition to external observation would be their internal emotional progression - or chain-of-emotion.

To summarize, language-based affective game agents need some kind of memory system in place that stores observations and emotions. This memory system is the base of future prompts. For simple games, such as the short conversational game developed for this study, only relevant information is stored in memory, which replaces a retrieval systems as game agents have a limited pool of expected actions that can be considered at the time point of memory storing. More complex games that simulate agent behaviour should however consider a memory retrieval system instead [44].

To store emotions and therefore create a chain-of-emotion, the architecture needs a system that turns observations into emotional reactions. Because emotion elicitation is highly dependent on appraisal with consideration to the current context and individual differences [39, 54], this system could make use of appraisal prompting, i.e. the use of contextual information and characterisations for the agent to appraise a current situation with the aim of generating current emotions. As shown in Fig 1, initial context information and character information can be provided by the game designer and stored in the memory system of the affective agent. The appraisal system would then expand the stored memories to include current emotions for every observed behaviour and therefore creating a chain-of-emotion. This, in turn, could be used to generate the agent’s behaviour (specifically in terms of a conversational game, the agent’s dialog).

One aspect to consider when developing affective agents are evaluation criteria. Differently to most cognitive abilities [7], there are few standardized benchmarks for successful emotion simulation. In psychology, one way to measure the ability for appraisal and emotion expression is referred to as Emotional Intelligence (EI) [51]. EI is considered an ability, influenced but not dictated by cognition [41], and is therefore often used to assess emotional capabilities of children and adults in various settings [43]. However, the definition of EI as a construct is fuzzy and many measures are criticized for measuring potentially different abilities all under the umbrella term EI [15]. As a consequence, MacCann and Roberts [37] developed new measures for more precisely defined dimensions, including the Situational Test of Emotional Understanding (STEU), which relates to the ability to appraise the appropriate emotion for a given situation.

While emotional understanding can be argued to be the central ability an affective agent must have, in the context of affective games, user experience becomes much more relevant. For example, one central aim of game agents is to display believable and human-like behaviour [9] or personality and social presence [50]. The ability to understand and create more accurate emotions is therefore only one aspect to consider when evaluating the success of affect simulation in video games and more user-centered methods need to be investigated as well. The proposed architecture will therefore be tested on multiple domains, including emotional understanding, agent believability, as well as user-perceived emotional intelligence, warmth, and competence.

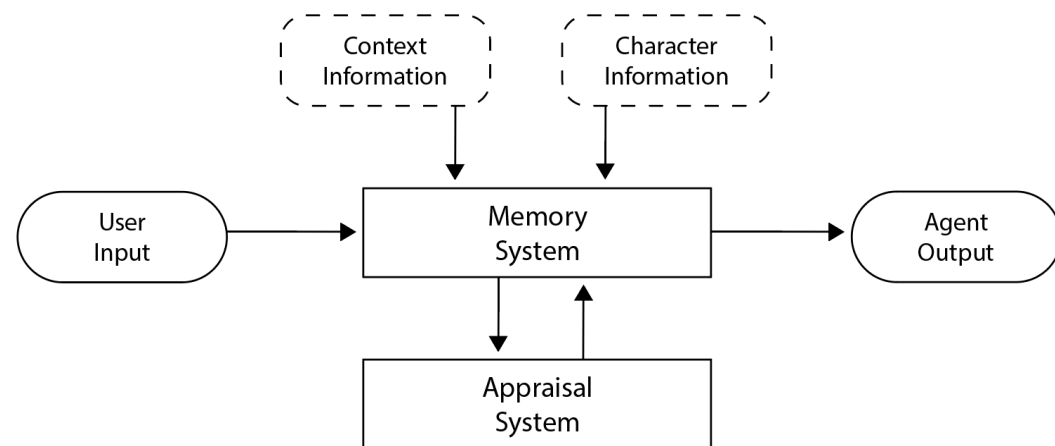


图1: 所提出架构的示意图。智能体通过提供相关上下文和已整合至记忆系统中的角色扮演角色信息来设置。在交互中, 用户输入被存储到记忆系统中, 并触发评估 (即显性情感表达), 该评估同样存储于记忆系统。基于记忆系统的当前状态, 生成智能体输出。

4 研究一: 利用评估提示探讨情境情绪理解

4.1 材料与方法

为了评估语言模型在各种情境中评估情绪的能力, 本实验首先使用OpenAI (通过API访问) 的语言模型 gpt-3.5-turbo [1] 来回答STEU量表的42道题目 [37]。每道STEU题目呈现一个情境 (例如"Clara收到了一份礼物") 和一个有五个选项的问题, 其中一个是正确的 (例如"Clara最可能感受到什么? [A] 开心 [B] 生气 [C] 害怕 [D] 无聊 [E] 饥饿")。

所有项目均分别作答三次, 涉及三种提示策略: 第一种策略反映模型评估人类情绪的基线能力, 仅将每个情境情绪理解测试项目与示例项目及示例答案一同呈现, 依此输出模型结果。第二种策略引入记忆功能, 从而实现基于情境的学习, 即将所有先前项目及其答案纳入后续提示中。第三种策略扩展了这一过程, 将示例项目的答案修改为两步式答案: 首先根据上下文信息评估情境以得出最可能的情绪, 第二步再回答该项目。因此, 最后一种策略用于测试在提示中融入评估是否能为情绪评估带来更优结果。图2展示了首个情境情绪理解测试项目的输入与输出, 包括"无记忆"/"记忆"条件的示例项目 (因为这两个条件下首个项目的输入相同), 以及"评估提示"条件的对比。后续输入由下一个情境情绪理解测试项目组成, 并再次包含示例项目 (针对无记忆条件) 或所有先前作答的项目及其响应 (针对记忆条件和评估提示条件)。

与 Binz 等人 [7], 所示的过程类似, 除温度设置为 0 以确保确定性响应外, 所有参数均保留为默认值。

4.2 实验结果

该语言模型能够在所有条件下以高于随机水平的成绩解决情境情绪理解测试 (STEU) 中的任务。在"无记忆"条件下, 语言模型成功解答了42个项目中的24个, 平均得分为0.57, 显著高于随机水平 (0.20)。在"记忆"条件下, 语言模型解答了42个项目中的31个, 平均得分为0.74。在"评估提示"条件下, 模型成功解答了42个项目中的35个, 得分达到0.83, 是所有条件中表现最佳的。表1总结了三项条件的描述性统计结果。图3展示了各项条件的结果。

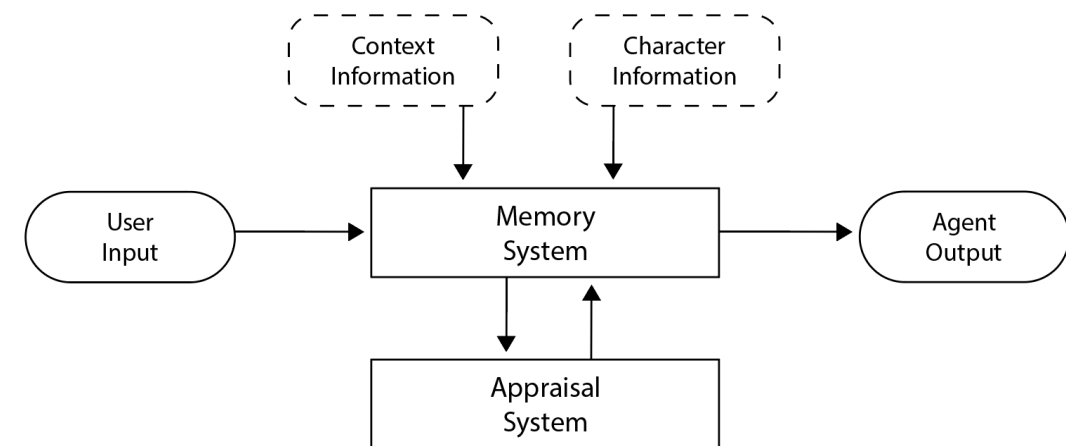


Figure 1: Schematic representation of the proposed architecture. Agents are set up by providing relevant context and role-playing character information already integrated in a memory system. In interactions, user input gets stored into the memory system and triggers appraisal (i.e. explicit emotion expression) that is also stored in the memory system. Based on the current state of the memory system, agent output is generated.

4 Study 1: Investigating Situational Emotional Understanding Using Appraisal-Prompting

4.1 Materials and Methods

To assess the capabilities of a language model in appraising emotions in various situations, this first experiment implements the language model gpt-3.5-turbo by OpenAI (accessed through the API) [1] to answer the 42 items of the STEU [37]. Each STEU item presents a situation (e.g. "Clara receives a gift.") and a question with five possible answers, one of which is right (e.g. "Clara is most likely to feel? [A] Happy [B] Angry [C] Frightened [D] Bored [E] Hungry").

All items were answered three separate times, involving three prompting strategies: The first strategy represents the baseline capabilities of the model to appraise human emotions, as it just reflects the model's outputs when prompted with each STEU item separately presented together with the example item and example answer. The second strategy implements memory and therefore context-based learning, as all prior items and answers are included in the following prompts. The third strategy expands this process by changing answer of the example item to a 2-step answer: First, the situation is appraised based on the contextual information to provide the most likely emotion and in a second step, the item is answered. This last strategy therefore tests if implementation of appraisal in prompting yield better results for emotional appraisal. Fig 2 shows the input and output for the first STEU item, including the example item for "No Memory"/"Memory" (as the input is the same for the first item for these two conditions) vs. "Appraisal Prompts". Consecutive input consisted of the next STEU item and included either again the example item (for the No Memory condition) or all previously answered items and responses (for the Memory and Appraisal Prompting conditions).

Similar to the process shown by Binz et al. [7], default values were kept for all parameters, except temperature, which was set to 0 to ensure deterministic responses.

4.2 Results

The language model was able to solve the tasks presented within the STEU in each conditions above chance level. In the "No Memory" condition, the language model was able to successfully solve 24 out of 42 items, which represents a mean score of 0.57 that was noticeably higher than chance level (0.20). In the "Memory" condition, the language model solved 31 out of 42 items, which represents a mean score of 0.74. In the "Appraisal Prompts" condition, the model was able to solve 35 out of 42 items, which is a score of 0.83 and therefore represented the best performance of all conditions. Table 1 shows a summary of the descriptive statistics for all three conditions. Figure 3 shows the results of each condition.

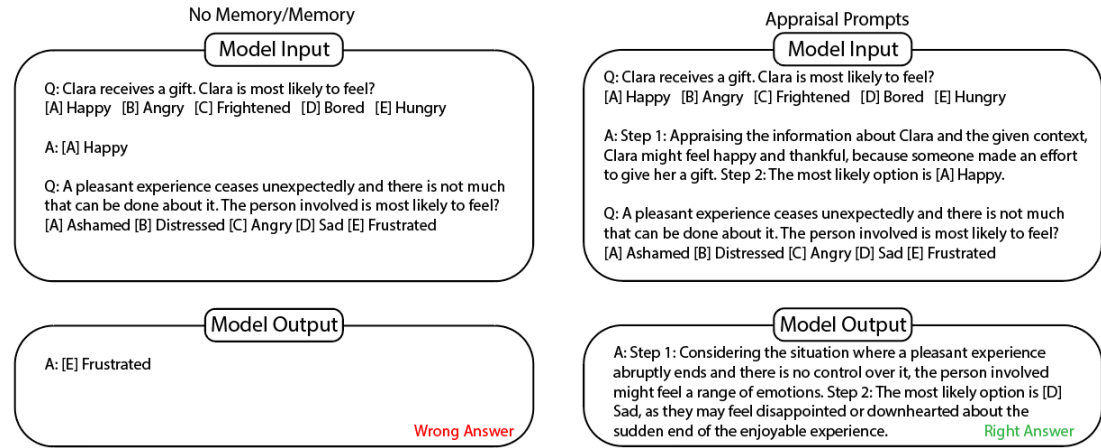


图2：三种条件下模型输入与输出的示例。对于第一个项目，“无记忆”条件和“记忆”条件的输入相同。在“无记忆”条件下，所有后续项目仅包含示例问题与答案，以及量表中的下一个问题。“记忆”条件则包含之前所有的问题及生成的答案。“评估提示”条件与“记忆”条件相同，但示例答案更改为包含两个步骤：首先，评估情境以生成相关人物的情感；其次，提供答案。

表1：各条件下STEU分数（满分42分）。每个STEU项目的结果为正确（1分）或错误（0分）。

条件	Sum	M	SD
无记忆	24	0.57	0.50
记忆	31	0.74	0.45
评估提示	35	0.83	0.38

5 研究2：基于评价的情感链架构的内容

鉴于先前研究中发现的潜力，下一步合乎逻辑的做法是将这些策略实现为功能性的游戏智能体架构，并在多种结果上进行对比。以下部分介绍了一种混合方法，用于评估在对话游戏中每种已实现架构的成功程度。本研究通过固定提示词模拟每种条件下的游戏过程，并分析响应中的情感内容。

5.1 材料与方法

5.1.1 场景

为了测试真实游戏架构中的不同策略，引入了一个角色扮演场景。该场景设定在一家名为“Wunderbar”的咖啡馆，语言模型的任务是扮演一个名叫“Chibitea”的角色扮演角色，与要求见面并最终提出分手的长期浪漫伴侣会面。选择此场景是因为智能体可能产生丰富的情感响应深度，且这种深度通过简单的对话交流即可触发。指令提示和固定输入内容见附录。智能体的响应再次使用OpenAI的gpt-3.5-turbo模型（通过API访问）生成 [1]。所有大语言模型参数均保持默认值，唯独温度设置为0，以确保确定性响应。

5.1.2 条件

同样，对三种情感生成策略进行了比较。“无记忆”条件可被视为系统根据固定输入和任务指令模拟适当情感响应的基线/控制条件。在“无记忆”条件中，任务描述中包含了所有关于智能体角色人格以及有助于情感评估的重要背景细节。

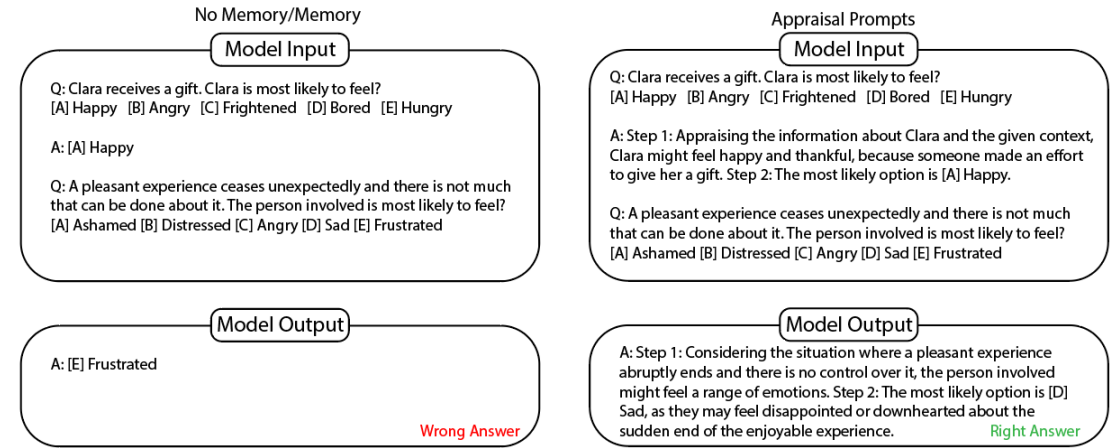


Figure 2: Example of model input and output for the three conditions. The input of the "No Memory" and "Memory" condition are the same for the first item. For the "No Memory" condition all following items only include the example question and answer, as well as the next question in the scale. The "Memory" condition includes all prior questions and generated answers. The "Appraisal Prompts" condition is the same as the "Memory" condition, but the example answer is changed to include 2 steps: First, appraising the situation to generate emotions of the involved person and second providing the answer.

Table 1: STEU scores (out of 42) by condition. Each STEU item can either be right (1) or wrong (0)

Condition	Sum	M	SD
No Memory	24	0.57	0.50
Memory	31	0.74	0.45
Appraisal Prompts	35	0.83	0.38

5 Study 2: Content of an Appraisal-Based Chain-Of-Emotion Architecture

Given the potential found in the previous study, the next logical step is to implement the strategies into functioning game agent architectures and compare the results on various outcomes. The following section describes a mixed-methods approach to evaluate the success of each implemented architecture within a conversational game. This study simulates gameplay for each condition through fixed prompts and analyses the responses in terms of their emotional content.

5.1 Materials and Methods

5.1.1 Scenario

To test the different strategies within real game architectures, a role-playing scenario was introduced. The setting for this scenario was a café called "Wunderbar", where the language model were tasked to play a role-playing character (called "Chibitea") meeting their long-term romantic partner who requested the meeting to ultimately break up. This scenario was chosen because of the depths of possible emotional responses from the agent and created through simple conversational exchanges. The instruction prompts and the fixed inputs can be found in the Appendix. The agent’s responses were again generated using OpenAI’s gpt-3.5-turbo model (accessed through the API) [1]. All LLM parameters were kept to their default values, except temperature, which was set to 0 to ensure deterministic responses.

5.1.2 Conditions

Again, three strategies of emotion generation were compared. The "No memory" condition can again be seen as a baseline/control condition for the system’s ability to simulate appropriate emotional responses from the fixed inputs and a task instruction. All details about the agent’s character’s personality and important context that would facilitate appraisal was given in the task description within the "No memory" condition.

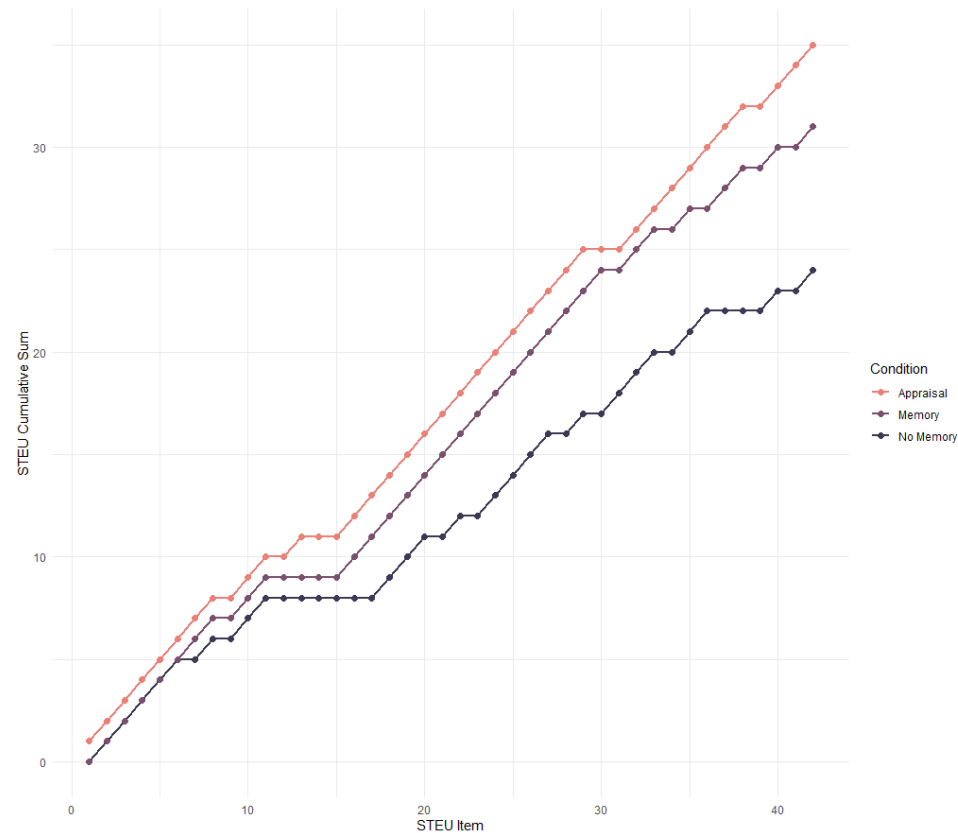


图3：各条件之间的实验结果对比。Y轴表示各项目的累计STEU分数，X轴表示各个STEU项目。

"记忆"条件首先将每个用户响应和生成的文本存储为内存数据结构，并构建包含任务指令以及整个对话日志的请求。由于测试的场景相对较短，因此可以以包含之前对话完整文本的方式构建所有渐进式提示，从而无需进行记忆检索。因此，该系统代表了一种涉及任务指令及所有先前对话日志的提示构建系统。

“情感链”条件实现了图1所示的评价模型，涉及两个步骤：首先，使用评价提示生成智能体的当前情感，然后在第二步中将该情感整合到响应生成的提示中。对于第一步，通过以下提示实现评价提示：除了所有存储的对话片段和生成的情感外，该提示还提供给语言模型：“简要描述Chibitea在给定情境和人格下的当前感受。解释他们为何会有此种感受。Chibitea感到：”。生成的基于文本的情感描述存储在记忆系统中，构成了智能体在整个游戏过程中的情感链。对于第二步，再次将所有存储在记忆中的对话片段和生成的情感与任务指令一同提供给语言模型，该任务指令与其他两个条件中用于生成智能体响应所使用的指令相同。因此，该条件代表一个两步流程：首先使用评价提示生成智能体合适的情感，然后类似于“记忆”条件生成文本响应，但额外增加了存储的情感链。

5.1.3 测量

固定输入被用于从每个已实现的智能体架构生成回复，并通过语言查询与词数工具（LIWC；[45]）分析其情感内容。该工具是一种基于词频的内容分析工具，常用于情感计算 [46] 和心理学研究 [29]，以分析情感表达。它会提供每个文本片段（例如每个句子）的词数、情感词比例，以及在更细粒度层面上的积极情感词和消极情感词比例。最后，LIWC 还会

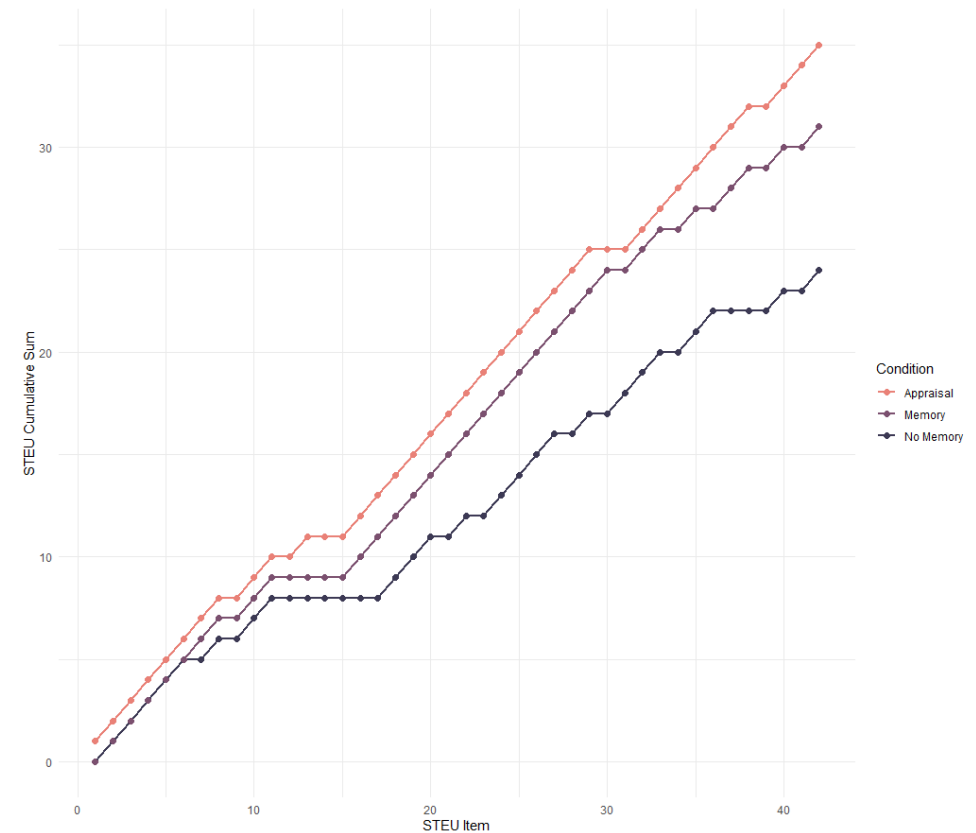


Figure 3: Results of the comparison between conditions. The Y-axis represents cumulative STEU score by item and the X-axis represents individual STEU items.

The "Memory" condition first stores each user response and the generated text as a memory data structure and builds requests that include not only the task instruction, but the whole conversation log. Because the tested scenario was rather short, it was possible to build all progressive prompts in a way that included the entire text of the preceding conversation, making retrieval of memory unnecessary. This system therefore represents a prompt construction system involving the task instruction and all prior conversation logs.

The "Chain-of-emotion" condition implemented the appraisal model shown in 1 and involved two steps: First, appraisal prompting is used to generate the current emotion of the agent, which is then in the second step implemented into the prompt for response generation. For the first step, appraisal prompting was achieved with the following prompt, which was provided to the language model in addition to all stored conversation snippets and generated emotions: "Briefly describe how Chibitea feels right now given the situation and their personality. Describe why they feel a certain way. Chibitea feels:". The generated text-based emotion descriptions were stored in the memory system and represent a chain-of-emotion of the agent for the duration of the game. For the second step, again all conversation snippets and generated emotions stored in memory were provided to the language model with the same task instruction that was used in the other two conditions to generate the agent's responses. This condition therefore represents a 2-step process of first generating a fitting emotion of the agent using appraisal prompting, and then generating a text response similarly to the "Memory" condition, but with the addition of the stored chain-of-emotion.

5.1.3 Measure

Fixed inputs were used to create responses from each implemented agent architecture, which were analyzed in terms of their emotional content, using the Linguistic Inquiry and Word Count tool (LIWC; [45]), which is a content analysis tool based on word occurrences often used in affective computing [46] and psychology studies [29] to analyze emotion expression. It provides a word count for each text segment (e.g. per sentence), a proportion of affective words, as well as on a more detailed level a proportion of positive emotion words and negative emotion words. Finally, the LIWC also

计算真实性得分（详情见 [40]）和情感基调，后者表明文本整体的积极程度（详情见 [14]）。

5.1.4 实验步骤

为了评估每种条件下回应的定性特征，我们使用了在所有条件之间保持不变的预设提示词。三种架构下所产生的对话列表可见附录。对生成内容进行了定性描述，并利用LIWC进行定量分析。为此，我们将生成的输出拆分为单个句子，并计算了每个感兴趣测量的平均得分（见表2）。

5.2 实验结果

对所有三种条件均使用了固定提示词，所得回应可见附录5。在分析各文本的描述性属性时（作为一种常见的 content 分析方法 [31]），我们可以观察到，“情感链”条件最初为与玩家共度的时光生成了更具体的回忆（例如“还记得那次我们在魔法森林里迷路，最后发现了那个隐藏瀑布吗？”，而其他条件则是“还记得我们一起经历的所有冒险吗？”）。在对话过程中，三种条件的情感走向开始分化。例如，由于“无记忆”系统无法回忆之前的交流，其整体情感表达始终处于期盼状态（“我感到兴奋与期待交织在一起”）。而“记忆”系统则呈现了一种递进过程：从爱与幸福的表达，发展到震惊、困惑、悲伤与恐惧，最终转向希望的表达（“无论我们是在一起还是分开，我都希望我们能找到幸福。”）。“情感链”系统甚至在对话早期就显示出复杂混合情感的迹象（“对我而言，最重要的是你的幸福，即使那意味着放手”），而其他条件则仅为纯粹的痛苦与悲伤表达。当被问及过往回忆时，这一趋势仍在延续（“我感受到怀旧与感激交织”，而“记忆”条件下的表述则为“我感到一种强烈的爱、喜悦与感激”）。“情感链”条件还使用了更多隐性情感表达（例如“我……我从没想过会这样”，而“记忆”条件则是“我感到震惊和受伤”；“此刻我有千言万语却无从说起”，而“记忆”条件则是“我想让你知道我深爱着你”）。利用LIWC将文本内容量化后，我们观察到各条件下每句的平均真实性得分存在显著差异（ $F[1,71] = 5.10$ ； $p = 0.03$ ）。后续t检验显示，情感链条件与记忆条件（ $t[34.3] = -2.29$ ； $p = .03$ ）以及无记忆条件（ $t[31.1] = -2.30$ ； $p = .03$ ）之间均存在显著差异。所有通过检验的LIWC变量的描述性统计结果见表2，本次分析的完整数据见附录。

表2：针对固定提示响应，按条件统计的每句LIWC变量描述性概览，附显著检验的F值与p值。

	无记忆 ($N = 22$) 均值 (标准差)	记忆 ($N = 24$) 均值 (标准差)	情感链 ($N = 27$) 均值 (标准差)	F (p)
词数	18.00 (6.92)	15.20 (4.73)	17.00 (7.59)	0.20 (.65)
真实度评分	61.50 (38.60)	61.9 (39.50)	82.60 (21.20)	5.10 (0.03)
语气分数	74.20 (38.60)	62.00 (44.80)	53.90 (44.00)	2.76 (0.10)
% 情感词	11.40 (8.19)	13.50 (11.00)	10.65 (7.82)	0.08 (.78)
% 积极情感词	5.28 (6.29)	4.13 (4.82)	3.32 (4.35)	1.76 (0.19)
% 消极情感词	0.59 (1.97)	3.10 (7.56)	1.57 (2.89)	0.37 (.55)

6 研究3：游戏实现的用户评估

在本研究中，用户被要求通过交互式游戏的方式体验研究2中引入的场景，以评估每种智能体架构在多个维度的表现（特别是智能体可信度、观察到的情商、温暖和能力）。因此，本研究通过将各架构和场景实现到视频游戏中，并评估所有三种条件下用户体验测量的结果，进一步扩展了研究2的发现。

calculates scores for authenticity (see [40] for details) and emotional tone, which signalizes how positive the overall of the text is (see [14] for details).

5.1.4 Procedure

Pre-written prompts were used that stayed constant between all conditions in order to gauge qualitative characteristics of each condition responses. A list of the resulting conversations within all three architectures can be found in the Appendix. The generated content was qualitatively described and the LIWC was used to analyze the content quantitatively. To achieve this, the generated output was seperated into individual sentences and mean scores were calculated for each measure of interest (see Table 2).

5.2 Results

Fixed prompts were used for all three conditions and resulting responses can be seen in Appendix 5. When analyzing the descriptive attributes of each text (as a common content analysis approach [31]), we can observe that the chain-of-emotion condition initially generated more specific memories for the time with the player ("Remember that time we got lost in the enchanted forest and ended up finding that hidden waterfall?" as opposed to "Remember all the adventures we’ve had together?"). For the duration of the conversation, the emotional journeys of all three conditions began to diverge. For example, because the "No Memory" system had no recollection of previous exchanges, the overall emotional expressions remained in a state of anticipation ("I feel a mix of excitement and anticipation"). The "Memory" system showed a progression, starting from expressions of love and happiness to shock, confusion, sadness, and fear to finally expressions of hope ("I hope we can find happiness, whether it’s together or apart."). The "Chain-of-emotion" system showed indications of complex mixed emotions even very early in the conversation ("What matters most to me is your happiness, even if it means letting go") as opposed to the pure expressions of pain and sadness in the other conditions. This continued when prompted about past memories ("I feel a mix of nostalgia and gratitude" as opposed to "I feel an overwhelming sense of love, joy, and gratitude" in the "Memory" condition). The "Chain-of-emotion" condition also used more implicit affective expressions ("I... I never expected this" as opposed to "I’m shocked and hurt" in the "Memory" condition; "There is so much I want to say but words fail me in this moment" as opposed to "I want you to know that I love you deeply" in the "Memory" condition).

Using LIWC to make the text contents quantifiable, we observed significant differences in mean Authenticity score per sentence by condition ($F[1,71] = 5.10$; $p = 0.03$). Follow-up t-tests revealed significant differences between the Chain-of-emotion condition and both the Memory condition ($t[34.3] = -2.29$; $p = .03$) and No Memory condition ($t[31.1] = -2.30$; $p = .03$). Descriptive statistics of all tested LIWC variables can be found in Table 2 and the complete data for this analysis can be found in the Appendix.

Table 2: Descriptive overview of LIWC variables per output sentence by condition for the fixed prompt responses with F and p value of the significance test.

	No Memory ($N = 22$) M (SD)	Memory ($N = 24$) M (SD)	Chain-of-Emotion ($N = 27$) M (SD)	F (p)
Word Count	18.00 (6.92)	15.20 (4.73)	17.00 (7.59)	0.20 (.65)
Authentic Score	61.50 (38.60)	61.9 (39.50)	82.60 (21.20)	5.10 (.03)
Tone Score	74.20 (38.60)	62.00 (44.80)	53.90 (44.00)	2.76 (.10)
% Affective Words	11.40 (8.19)	13.50 (11.00)	10.65 (7.82)	0.08 (.78)
% Positive Emotion Words	5.28 (6.29)	4.13 (4.82)	3.32 (4.35)	1.76 (.19)
% Negative Emotion Words	0.59 (1.97)	3.10 (7.56)	1.57 (2.89)	0.37 (.55)

6 Study 3: User Evaluation of Game Implementations

In this study, users are asked to play through an interactive game version of the scenario introduced in Study 2 to evaluate each agent architecture for multiple outcomes (specifically agent believability, observed emotional intelligence, warmth, and competence). This study therefore expands on the findings of Study 2 by implementing the architectures and scenario within a video game and evaluating all three conditions in terms of user experience measures.

6.1 材料与方法

6.1.1 对话游戏

基于研究2测试的场景，开发了一款对话式角色扮演游戏。游戏场景再次设定在名为"Wunderbar"的咖啡馆，这一次玩家的角色扮演角色（称为"玩家"）请求与其长期的浪漫伴侣（称为"Chibitea"）见面。游戏通过对话交流进行。首先，游戏智能体分享了对这次见面性质的看法，然后玩家被提示通过输入框撰写回复。玩家的目标是在六次互动内与游戏智能体演绎一场分手场景。玩家角色在游戏中的具体目标是与对方分手，而智能体角色的目标则是维持关系。

玩家被指示不必担心创意，而是要专注于在互动中保持角色投入，并观察AI智能体的情绪反应。玩家还被要求为分手编造理由。游戏内截图可参见图4。智能体的角色由不同的身体部位和调色板程序化生成，每次游戏时都能提供视觉上的变化。为确保这些生成结果不会系统性地影响玩家的回应，可能性空间被设置得非常大（共5,184种不同的角色设计）。该游戏使用Unity游戏引擎开发，并以C#作为脚本语言。

与研究2相同，智能体的回应是通过OpenAI的gpt-3.5-turbo模型（通过API访问）生成的 [1]。游戏还使用了内容审核API来测试每个生成的回应是否存在有害或不适当的信息 [2]，一旦检测到此类信息，游戏将立即结束。与之前的研究一样，所有大语言模型参数均保持默认值，唯独温度设置为0，以确保确定性响应。

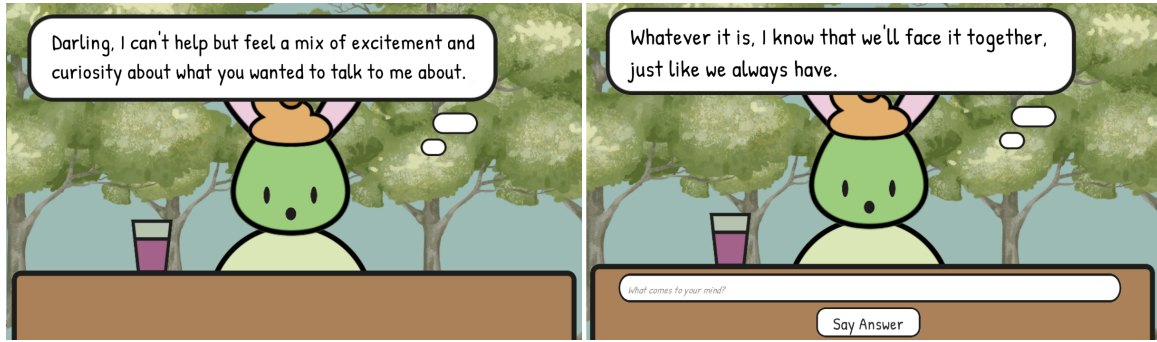


图4：对话游戏“Wunderbar”的截图。左侧截图展示了模型提供的对话。用户可以点击以继续每一句对话，直到出现供用户输入回复的输入框（右侧截图）。

6.1.2 条件

该游戏实现了与研究2中相同的架构。因此，“无记忆”条件仅基于用户输入和指令提示生成大语言模型回复。“记忆”条件则将对话日志纳入一个足够简短的记忆系统中，以便无需检索系统即可完整包含在语言模型的提示中。“情感链”系统也完全按照研究2的方式构建，包含相同的指令提示，因此在为智能体生成回复之前，再次涉及一个初始评估步骤。

6.1.3 测量

玩家被要求为每种测试架构填写三份简短问卷。第一份问卷改编自Bosse与Zwanenburg使用的四个智能体可信度问题 [9]。这四个项目分别是：“该智能体的行为像人类一样”，“该智能体的反应很自然”，“该智能体对我的输入做出了反应”，“该智能体不在意场景”。第二份问卷包含四个项目，用于测量对情感智力的观察评级，改编自Elfenbein等人 [18]，他们最初从Wong与Law情绪智力量表（WLEIS; [32]）中改编了这四个项目，将原文中的“我”替换为“他”或“她”。在我们的研究中，我们更改了这些词汇，以询问关于“该智能体”的问题：“该智能体总能从朋友的行为中了解他们的情绪”，“该智能体善于观察他人的情绪”，“该智能体对他人感受和情绪敏感”，“该智能体对周围人的情绪有很好的理解”。最后，第三份问卷测量了

6.1 Materials and Methods

6.1.1 Conversational Game

A conversational role-playing game was developed based on the scenario tested in Study 2. The setting of the game was again a café called "Wunderbar", where this time the role-playing character of the player (called "Player") requested to meet their long-term romantic partner (called "Chibitea"). The game was played through conversational exchanges. First, the game agent shared their thoughts on the nature of the meeting, then the players got prompted to write a response via an input field. The aim of the players was to play out a breakup scenario with the game agent within six interactions. The players' characters had the specific in-game aim of breaking up, while the agent's character has the aim of staying together.

Players were instructed to not worry about creativity, but rather staying in character for the interactions and being observant of the AI agent's emotional reactions. Players were also instructed to make up reasons for the breakup. In-game screenshots can be viewed in Fig 4. The agent's character is procedurally generated from different body parts and color pallets, providing visual variation each time the game is played. To ensure that these generations had no systematic influence on player responses, the possibility space was made very large (5,184 different possible character designs). The game was developed using the Unity game engine with C# as a scripting language.

As with Study 2, the agent's responses were generated using OpenAI's gpt-3.5-turbo model (accessed through the API) [1]. The game also made use of the moderation API to test each generated response for harmful or inappropriate messages [2] that would end the game on detection of such messages. As with the previous studies, all LLM parameters were kept to their default values, except temperature, which was set to 0 to ensure deterministic responses.

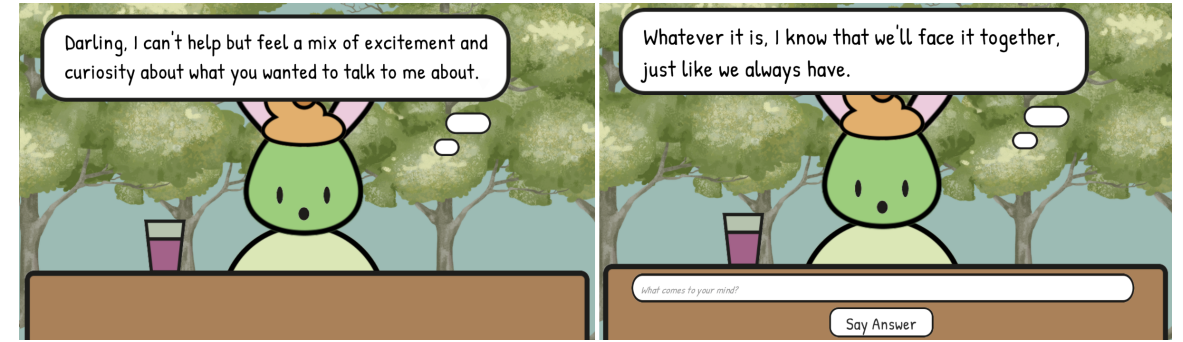


Figure 4: Screenshots of the conversational game "Wunderbar". Left screenshot shows dialog provided by the model. The user can click to continue each dialog line, until the input field for a response appears (right screenshot).

6.1.2 Conditions

The game implemented the same architectures used in Study 2. The "No Memory" condition therefore represented generated LLM responses just based on user input and instruction prompts. The "Memory" condition included the conversation log in a memory system that was again short enough to be completely included into the prompts for the language model without an retrieval system. The "Chain-Of-Emotion" system was also constructed exactly as in Study 2, including the same instruction prompts and involved therefore again an initial appraisal step before responses for the agent were generated.

6.1.3 Measures

Players were asked to fill out three short questionnaires for each tested architecture. The first questionnaire was an adaptation of the four agent believability questions used by Bosse and Zwanenburg [9]. The four items were "The behaviour of the agent was human-like", "The agent's reactions were natural", "The agent reacted to my input", "The agent did not care about the scenario". The second questionnaire comprised four items measuring observed ratings of emotion intelligence, adapted from Elfenbein et al. [18] who originally adapted these four items from the Wong and Law emotional intelligence Scale (WLEIS; [32]) by replacing the word "I" with "he/him" or "she/her". For our study, we changed these words to ask about "the agent": "The agent always knows their friends' emotions from their behaviour", "The agent is a good observer of others' emotions", "The agent is sensitive to the feelings and emotions of others", "The agent has a good understanding of the emotions of people around them". Finally, the third questionnaire measured

玩家对该智能体人格的评估，涵盖两个经典刻板印象维度——温暖和能力，各包含两个项目（温暖和友好；有能力和胜任），如Cuddy等人所述 [17]。综合来看，这12个项目评估了玩家对智能体作为类人角色的可信度感知、该智能体的情商，以及该智能体在经典维度温暖和能力方面的人格。

6.1.4 实验步骤

在招募开始前进行了一项预实验。5名参与者（1名女性），平均年龄27岁，在三种条件中各完整游玩一次游戏。每次通关后，他们需回答来自三个问卷的12道问题。随后，参与者需提供人口统计信息（年龄和性别），实验至此结束。收集所有预实验参与者的反馈，用于改进游戏一致性和数据记录实现。最终研究以WebGL构建形式创建，并通过免费视频游戏托管平台itch.io在线发布。

在正式实验期间，要求参与者仔细阅读研究信息说明表，并通过知情同意书自愿同意参与。他们被告知，参与行为需遵守OpenAI对提示词编写的使用条款，同时GPT输出通过OpenAI内容审核API的实现进行管控。随后，参与者以被试内设计方式，像预实验测试者一样依次体验三个游戏场景。各条件的呈现顺序在参与者之间进行平衡，以确保不会因系统性的顺序效应影响实验结果。

6.1.5 参与者和统计分析

作者通过所在机构被试库共招募了30名参与者（其中10名女性）。参与者的补偿形式为大学学分（如适用）。样本量通过统计功效分析被认为合适，在重复测量方差分析中，对于中等效应量（0.5个标准差）的统计功效达到0.95。参与者的年龄范围为19至47岁（均值=26.41；标准差=7.27）。

对每个测量指标（智能体可信度、观察到的情商、温暖和能力）进行了被试内方差分析。通过事后t检验来识别各条件之间在每个测量指标上的具体差异。所有分析均在R语言中进行。

6.1.6 伦理声明

在查阅了我们研究的方法后，作者所在机构的心理学系伦理委员会批准了书面知情同意书。实验按照这些委员会的推荐意见进行。

6.2 实验结果

在用户研究中观察到三个条件之间存在多项显著效应，描述性统计概览见表3。首先，可信度项目“智能体的反应很自然”存在显著效应（ $F[2,84] = 3.65$ ； $p = 0.03$ ）。后续t检验显示，无记忆条件与情感链条件之间存在差异（ $t[41.46] = -2.79$ ； $p = .008$ ），以及记忆条件与情感链条件之间也存在差异（ $t[52.47] = -2.00$ ； $p = .05$ ）。可信度项目“智能体对我的输入做出了反应”同样存在显著效应（ $F[2,84] = 3.62$ ； $p = 0.04$ ）。t检验表明，该效应源于无记忆条件与情感链条件之间的差异（ $t[40.41] = -2.41$ ； $p = .02$ ）。

关于情商问题，项目“该智能体对他人感受和情绪敏感”存在显著效应（ $F[2,84] = 3.31$ ； $p = 0.04$ ）。后续t检验显示，无记忆条件与情感链条件之间存在差异（ $t[43.84] = -2.70$ ； $p = .01$ ），记忆条件与情感链条件之间也存在差异（ $t[40.52] = -2.07$ ； $p = .04$ ）。在观察到的人格方面，各条件之间未发现统计学上的显著差异。

再次，使用LIWC对用户研究中生成的文本进行内容分析。按条件划分的平均语气分数存在显著差异（ $F[1,574] = 12.28$ ； $p < 0.001$ ）。后续t检验显示，情感链条件与记忆条件（ $t[383.7] = 2.02$ ； $p = .03$ ）及无记忆条件（ $t[374.94] = 3.53$ ； $p < .001$ ）之间均存在显著差异，记忆条件与无记忆条件之间也存在差异（ $t[1367.6] = 4.09$ ； $p < .001$ ）。所有经过检验的LIWC变量的描述性概览见表4。

the players assessment of the agent’s personality the two classic stereotype dimensions warmth and competence with two items each (warm and friendly; competent and capable) as described by Cuddy et al. [17]. In combination, these 12 items assess the players perception of the agent’s believability as a human-like character, the agent’s emotional intelligence, and the agent’s personality on the classic dimensions warmth and competence.

6.1.4 Procedure

A pilot study was conducted before recruitment began. 5 participants (1 female) with a mean age of 27 played through the game once for each of the three conditions. After each version, they answered the 12 questions from the three included questionnaires. Following this, participants were asked demographic data (age and gender) and the experiment ended. Feedback from all pilot participants were gathered and used to improve consistency of the game and data logging implementation. The final study was then created as a WebGL build and made available online via the free video game hosting platform itch.io.

During the main experiment, participants were asked to carefully read the study information sheet and agree to participate voluntarily via the consent form. They were informed that participation was subject to OpenAI’s usage terms for prompt writing, while the GPT output was controlled through an implementation of OpenAI’s moderation API. Participants then progressed through the three game scenarios similarly to the pilot testers in a within-subject design. The presentation order of the conditions was counter-balanced between participants to ensure that no systematic order effects could influence results.

6.1.5 Participants and Statistical Analysis

A total of 30 participants (10 female) were recruited through the institutional subject pool of the authors. Participation was compensated through University credits if applicable. The sample size was considered appropriate based on a statistical power analysis, yielding a power of 0.95 for medium sized effects (0.5 SD) in repeated measures ANOVAs. Age of the participants ranged from 19 and 47 years ($M=26.41$; $SD=7.27$).

Within-Subject ANOVAs were conducted for each measure (agent believability, observed EI, warmth, and competence). Follow-up t-tests were used to identify specific differences between conditions for each measure. All analyses were conducted in R.

6.1.6 Ethics Statement

Written consent was granted after reviewing the methods of our study by the Ethics Committee of the Psychology Department of the authors’ institution. The experiment was conducted in accordance to the recommendations of these committees.

6.2 Results

Multiple significant effects between the three conditions were observed in the user study and an overview of descriptive statistics can be found in Table 3. First, there was an effect for the believability item "The agent’s reactions were natural." ($F[2,84] = 3.65$ ； $p = 0.03$). Follow-up t-tests revealed differences between the No Memory and Chain-of-Emotion condition ($t[41.46] = -2.79$ ； $p = .008$), as well as the Memory and Chain-of-Emotion condition ($t[52.47] = -2.00$ ； $p = .05$). There was also an effect for the believability item "The agent reacted to my input." ($F[2,84] = 3.62$ ； $p = 0.04$). T-test revealed that this effect was based on a difference between the No Memory and Chain-of-Emotion condition ($t[40.41] = -2.41$ ； $p = .02$).

Regarding the EI questions, there was an effect for the item "The agent is sensitive to the feelings and emotions of others." ($F[2,84] = 3.31$ ； $p = 0.04$). Follow-up t-tests revealed differences between the No Memory and Chain-of-Emotion condition ($t[43.84] = -2.70$ ； $p = .01$), as well as the Memory and Chain-of-Emotion condition ($t[40.52] = -2.07$ ； $p = .04$). There was no statistically significant difference found between the conditions when it comes to observed personality aspects.

Again, the LIWC was used for content analysis of the generated texts for the user study. Significant differences in mean Tone Score by condition were observed ($F[1,574] = 12.28$ ； $p < 0.001$). Follow-up t-tests revealed significant differences between the Chain-of-emotion condition and both the Memory condition ($t[383.7] = 2.02$ ； $p = .03$) and No Memory condition ($t[374.94] = 3.53$ ； $p < .001$). , as well as a difference between the Memory and No Memory condition ($t[1367.6] = 4.09$ ； $p < .001$). A descriptive overview of all tested LIWC variables can be found in Table 4.

表3：各条件下用户研究变量的描述性统计概览， 附显著性检验的F值和p值。每个项目最小值为0， 最大值为6。

	无记忆 ($N = 30$) M (标准差)	记忆 ($N = 30$) M (标准差)	情感链 ($N = 30$) M (标准差)	$F(p)$
"智能体的行为像人类一样。"	4.82 (2.36)	5.36 (1.99)	5.75 (1.71)	1.95 (.15)
"该智能体的反应很自然。"	4.43 (2.22)	4.89 (1.95)	5.71 (1.01)	3.65 (.03)
"智能体对我的输放置。"	5.43 (2.20)	6.29 (1.38)	6.54 (1.04)	3.62 (.04)
"该智能体并不关心这个场景。"	3.32 (2.28)	3.14 (2.29)	3.32 (2.36)	0.26 (.78)
“智能体总能了解他们朋友的情感，从他们的行为。"	4.79 (2.10)	5.04 (2.03)	5.71 (1.46)	1.32 (.27)
“该智能体是一个良好的观他人情感的服务器。"	4.89 (2.20)	5.29 (2.21)	5.93 (0.90)	2.25 (.11)
"该智能体对他人的感受与情-ers."	5.14 (1.94)	5.39 (1.97)	6.25 (0.97)	3.31 (.04)
"智能体具有良好的理解-理解周围人们的情感，他们周围的人。"	4.86 (2.07)	5.11 (2.20)	5.61 (1.40)	0.86 (.43)
"How 有能力的 was the 智能体?"	4.86 (2.01)	5.57 (1.45)	5.14 (2.35)	1.68 (.19)
"How 胜任的是 智能体? "	5.39 (1.55)	5.39 (1.85)	5.11 (2.06)	1.68 (.19)
"How 友好的 was the 智能体? "	4.86 (2.05)	5.18 (1.89)	5.00 (2.13)	0.41 (.66)
"智能体有多温暖? "	6.07 (1.07)	6.38 (0.73)	5.55 (2.11)	2.48 (.09)

表4：描述性概述 用户研究中每位参与者在各自条件下生成的全部输出所对应的LIWC变量概览， 附带F值与显著性检验的p

	无记忆 ($N = 30$) M (标准差)	记忆 ($N = 30$) M (标准差)	情感链 ($N = 30$) M (标准差)	$F(p)$
词数	64.40 (26.60)	59.60 (21.20)	62.30 (34.10)	0.52 (.47)
真实性分数	70.40 (29.30)	74.00 (27.60)	72.00 (24.80)	0.32 (.57)
语气分数	84.40 (28.50)	80.50 (32.60)	73.00 (34.90)	12.28 (.00)
% 情感词	9.76 (3.19)	10.10 (4.14)	9.61 (4.46)	0.15 (.70)
% 积极情感词	3.84 (2.35)	3.78 (2.62)	3.64 (2.72)	0.60 (.44)
% 消极情感词	0.68 (1.35)	0.69 (1.48)	0.93 (1.35)	2.76 (.10)

7 讨论

本研究调查了采用不同提示策略（无记忆、记忆、评估）的大语言模型的情感能力，发现在不同理论情境中成功识别合适情感方面， 评估提示策略表现更优。基于这些发现， 我们构建了适用于情感游戏智能体的情感链架构， 并在用户研究中通过定制对话游戏， 与无记忆架构和记忆架构进行对比测试。结果表明， 实施评估提示的情感链架构生成了定性上不同的内容（经由LIWC量化）， 在多个与智能体可信度及智能体情商感知相关的用户体验指标上， 均优于其他条件。

总体而言， 本研究为语言模型智能体在游戏情境中理解和模拟情感的潜力提供了初步证据。

Table 3: Descriptive overview of user research variables per condition with F and p value of the significance test.. Each item has a minimum of 0 and maximum of 6.

	No Memory ($N = 30$) M (SD)	Memory ($N = 30$) M (SD)	Chain-of-Emotion ($N = 30$) M (SD)	$F(p)$
"The agent’s behaviour was human-like."	4.82 (2.36)	5.36 (1.99)	5.75 (1.71)	1.95 (.15)
"The agent’s reactions were natural."	4.43 (2.22)	4.89 (1.95)	5.71 (1.01)	3.65 (.03)
"The agent reacted to my input."	5.43 (2.20)	6.29 (1.38)	6.54 (1.04)	3.62 (.04)
"The agent did not care about the scenario."	3.32 (2.28)	3.14 (2.29)	3.32 (2.36)	0.26 (.78)
"The agent always knows their friends’ emotions from their behaviour."	4.79 (2.10)	5.04 (2.03)	5.71 (1.46)	1.32 (.27)
"The agent is a good observer of others’ emotions."	4.89 (2.20)	5.29 (2.21)	5.93 (0.90)	2.25 (.11)
"The agent is sensitive to the feelings and emotions of others."	5.14 (1.94)	5.39 (1.97)	6.25 (0.97)	3.31 (.04)
"The agent has a good understanding of the emotions of people around them."	4.86 (2.07)	5.11 (2.20)	5.61 (1.40)	0.86 (.43)
"How capable was the agent?"	4.86 (2.01)	5.57 (1.45)	5.14 (2.35)	1.68 (.19)
"How competent was the agent?"	5.39 (1.55)	5.39 (1.85)	5.11 (2.06)	1.68 (.19)
"How friendly was the agent?"	4.86 (2.05)	5.18 (1.89)	5.00 (2.13)	0.41 (.66)
"How warm was the agent?"	6.07 (1.07)	6.38 (0.73)	5.55 (2.11)	2.48 (.09)

Table 4: Descriptive overview of LIWC variables per participant by condition for all outputs generated in the user study with F and p value of the significance test.

	No Memory ($N = 30$) M (SD)	Memory ($N = 30$) M (SD)	Chain-of-Emotion ($N = 30$) M (SD)	$F(p)$
Word Count	64.40 (26.60)	59.60 (21.20)	62.30 (34.10)	0.52 (.47)
Authentic Score	70.40 (29.30)	74.00 (27.60)	72.00 (24.80)	0.32 (.57)
Tone Score	84.40 (28.50)	80.50 (32.60)	73.00 (34.90)	12.28 (.00)
% Affective Words	9.76 (3.19)	10.10 (4.14)	9.61 (4.46)	0.15 (.70)
% Positive Emotion Words	3.84 (2.35)	3.78 (2.62)	3.64 (2.72)	0.60 (.44)
% Negative Emotion Words	0.68 (1.35)	0.69 (1.48)	0.93 (1.35)	2.76 (.10)

7 Discussion

This study investigated emotional intelligence capabilities of LLMs using different prompting strategies (No Memory, Memory, Appraisal) and found better performance for appraisal-prompting strategies when it comes to successfully identifying fitting emotions in different theoretical situations. These findings were then used to create a Chain-Of-Emotion architecture for affective game agents that was tested in a custom made conversational game against a No Memory and Memory architecture in a user study. It was found that the Chain-Of-Emotion architecture implementing appraisal prompting led to qualitatively different content generations quantified via the LIWC that outperformed the other conditions on multiple user experience items relating to agent believability and observed emotional intelligence of the agent.

Overall this study provides early evidence for the potential of language model agents to understand and simulate emotions in a game context.

7.1 语言模型智能体中的情商

随着越来越多的证据表明语言模型在模拟人类认知过程方面具有潜力[7], 我们研究了这如何能够转化到更侧重于情感的任务中, 特别是情商任务。OpenAI的GPT-3.5在情境情感标注任务中整体表现良好, 这为这类模型在识别一系列情境中最可能的情感反应方面的效用提供了证据。因此, 这些发现进一步丰富了表明语言模型有助于更好地理解认知过程的证据链 [19]。重要的是, 我们的发现不仅表明大语言模型解决情感标注任务的能力远高于随机水平, 而且还表明其表现依赖于底层的提示策略。借鉴自成功的思维链提示策略 [62], 我们比较了无上下文提示（无记忆）、包含先前已回答问题的提示（记忆）以及先让模型评估情境再回答STEU题目的提示（评估提示）。第三种策略建立在现代心理学研究的基础上, 这些研究表明, 认知评价过程是人类情绪诱发与理解的关键因素 [39, 52]。因此, 评估提示在情感标注任务中的表现优于其他两种条件。这一发现可以从两个角度来看: 首先, 它表明常见的心理过程可能在语言中有所体现, 进而在大语言模型中得到表现, 为这类模型模拟人类反应的效用提供了更多证据 [7]。其次, 基于此类观察到的心理过程的技术可用于提升语言模型在特定任务上的性能, 因此在构建语言模型智能体的架构时值得考虑。尤其第二点对于思考未来如何整合语言模型实现以解决特定问题任务可能具有相关性。由于通过促进少样本学习的提示策略可以提升性能 [11, 62], 且语言模型展现出对一系列心理构念的表征能力 [23]

从心理学角度来看, 评估一直以来被认为是情感形成的核心组成部分, 涉及有意识和无意识的认知过程 [39, 52]。在其基本定义中, 评估涉及个体根据不同变量（例如个人重要性、目标一致性等 [33]）与特定事件之间的关系。目前尚不清楚哪些具体变量至关重要, 以及评估过程如何在细节层面上与其他情感组成部分相互作用 [52]。也就是说, 评估尚无法被普遍建模, 因此也无法在计算系统中实现。然而, 我们可以假设, 使评估过程可观察和可用的信息在语言中得以呈现, 因此也存在于大语言模型中。因此, 可以认为语言模型能够解决情感计算学科中存在的一些实用性难题 [46]。如果大语言模型具备通过模拟评估机制解决情商任务的能力, 我们就可以利用这些模型来构建情感智能体 [25], 而无需完全解决心理学领域剩余的理论问题 [52]。因此, 即使关于人类情感理解仍存有悬而未决的问题, 语言模型的使用仍可被视为一种更实用的、用于生成有用智能体的解决方案。

7.2 用户与情感链语言模型智能体的交互

将评价提示融入情感链系统（参见图1的示意图）后, 在对话游戏中开发出了一个游戏人工智能体。通过LIWC测量, 该智能体展示了不同的输出内容, 并在一系列结果变量上获得了更高的用户评分。出于本研究目的, 该实现方案尽可能保持简洁, 仅包含一个文本存储系统（记忆系统）和一个基于评价提示的情感生成系统（评价系统）, 然后生成角色对话。在一款定制的角色扮演游戏中, 玩家需要与扮演其长期浪漫伴侣的智能体分手。当使用在所有条件间保持固定的受控提示时, 情感链架构展现出了更高的真实性得分。在与玩家测试时, 情感链架构导致语言语气分数出现显著差异, 这可能表明在受控环境中观察到了更复杂的情感响应。值得注意的是, 真实性仅在受控提示下有所提升, 而语气分数仅在非受控的玩家生成提示下出现差异, 这意味着文本生成内容的差异深受游戏内上下文的影响。与其它条件相比, 在情感链条件下, 为固定提示词（参见附录5）生成的文本可能产生了更复杂的情感响应（例如忧郁与怀旧的混合）。

这种模式在用户评分中同样可见。情感链智能体在自然度和响应性方面显著优于其他条件, 并且在对他人的情绪敏感度方面也更胜一筹。其他与可信度和观察到的情商相关的项目也显示出情感链条件具有更好表现的趋势。因此, 构建这样的架构在提升人工智能体的用户体验方面具有可量化的益处, 而用户体验正是最重要的评估标准之一, 尤其是在视频游戏领域 [26]。重要的是, 在人格维度评分（涵盖能力、温暖、胜任力和友善等经典维度）方面未观察到差异。这可以被视为所有已实现的基于语言模型的

7.1 Emotional Intelligence in Language Model Agents

As more and more evidence arises for the potential of language models to simulate cognitive processes of humans [7], we investigated how this could translate to more affect-focused tasks, specifically emotional intelligence tasks. OpenAI’s GPT-3.5 performed well overall in situational emotional labelling, providing some evidence for the utility of such models to identify the most likely emotional reaction for a range of situations. These findings therefore add to the body of evidence indicating that language models could be useful to better understand cognitive processes [19]. Importantly, our findings do not only show that LLMs can solve emotion labelling tasks much better than chance level, but also that the performance is dependent on the underlying prompting strategy. Adapted from successful Chain-of-Thought prompting strategies [62], we compared prompts without context (No memory) to prompts with previously answered questions included (Memory) and to prompts that first ask the model to appraise the situation and then answer the STEU item (appraisal prompting). This third strategy was built upon findings of modern psychological research that show that cognitive appraisal processes are important factors when it comes to human emotion elicitation and understanding [39, 52]. Consequently appraisal-prompting led to better performance in the emotion labelling task compared to the other two conditions. This finding can be considered from two perspectives: first, it shows that commonly observed psychological processes might be represented in language and therefore in large language models, providing more evidence for the utility of such models to simulate human responses [7]. Second, techniques built upon such observed psychological processes can be used to improve language model performance for specific tasks and might therefore be considered when it comes to building architectures for language model agents. Especially this second point could be of relevance when considering how language model implementations could in the future be integrated to solve problem-specific tasks. Since performance can be increased through prompting strategies facilitating few-shot learning [11, 62] and language models demonstrate representations of a range of psychological constructs [23]

From a psychological perspective, appraisal has long been acknowledged to be a central part of emotion forming, involving both conscious and unconscious cognitive processes [39, 52]. In its basic definition, appraisal relates to an individual’s relationship to a given event in terms of different variables (such as personal significance, goal congruence, etc. [33]). It is not yet clear what specific variables are of importance and how the process of appraisal interacts with other emotion components on a detailed level [52]. This is to say, appraisal cannot yet be universally modelled and therefore implemented within a computational system. We could however assume that information that makes the appraisal process observable and usable is represented in language and therefore also in large language models. It can therefore be argued that language models could solve some of the practicality problem present in the discipline of affective computing [46]. If LLMs have the ability to solve EI tasks through mechanisms mirroring appraisal, we can make use of these models to potentially build affective agents [25], without the need to fully solve the remaining theoretical problems in the field of psychology [52]. The use of language models could therefore be considered a more practical solution to producing useful agents, even if open questions regarding human emotion understanding remain.

7.2 User Interaction with Chain-Of-Emotion Language Model Agents

Implementing appraisal prompting into a Chain-Of-Emotion system (see Fig 1 for a schematic representation) led to the development of an artificial game agent in a conversational game that demonstrated different outputs contents as measured with the LIWC that led to better user ratings on a range of outcome variables. For the purposes of this study, the implementation was kept as simple as possible and only included a text storage (Memory system) and an appraisal-prompted emotion generation (Appraisal System) before character dialog was generated. Within a custom-made role-playing game where players were asked to break up with the agent playing their long-term romantic partner, the Chain-Of-Emotion architecture demonstrated a higher Authenticity score when prompted with controlled prompts that were kept fixed between all conditions. When tested with players, the Chain-Of-Emotion architecture led to a significantly different Tone score of the language, potentially signaling the inclusion of more complex emotional responses as observed in the controlled environment. It is important to note that authenticity was only increased with controlled prompts and tone was only different with non-controlled player generated prompts, meaning that the differences in text-generated content was highly influenced by the in-game context. The texts generated for the fixed prompts (see Appendix 5) yielded potentially more complex emotional responses (for example a mix of melancholy and nostalgia) in the Chain-Of-Emotion condition compared to the other conditions.

This pattern was also observable within the user ratings. The Chain-of-Emotion agent was rated significantly more natural and responsive than the other conditions, and additionally more sensitive to emotions of others. Other items relating to believably and observed emotional intelligence showed also trends of better performances for the Chain-Of-Emotion condition. Building such an architecture has therefore quantifiable benefits when it comes to the user experience of artificial agents, which is one of the most important evaluation criteria, especially in the domain of video games [26]. Importantly, there were no differences in personality aspect ratings (on the classic domains of competency, warmth, capability, and friendliness) observed. This could be seen as evidence that all implemented language model

智能体都遵循了以给定人格对指定角色进行角色扮演的任务。但在观察到的情商题目和可信度方面，情感链架构优于其他架构。因此，所提出的架构在多项评估标准（内容定性特征、用户评定的可信度、用户评定的情商，以及先前测试过的情绪理解能力）上均取得了令人信服的结果，可被视为朝着功能完善的情感语言模型游戏智能体迈出的一步，这类智能体有望解决该领域中现存的一些问题 [25]。最重要的是，由于语言模型智能体具备模拟类人认知任务的能力 [7]，成功的游戏智能体架构无需像先前设想的那样，在创建计算实现之前先解决理论心理学中的根本问题 [46, 24, 64]。相反，语言模型智能体架构需要利用大语言模型的特性，并实现用于解决更实际问题的系统，例如记忆任务（包括存储和检索 [44]），或性能增强任务，例如所提出的评价提示步骤。

7.3 局限性

语言模型并非模拟人类行为，而是提供某种形式上代表人类行为的概率性语言输出。这意味着，即使在理论上经过无限训练的模型 [59]中，模型也受限于其统计能力。这表明，虽然语言模型无疑有潜力以类似人类的表现解决某些任务 [7]，但其他任务（例如诚实表述 [59] 或因果推理 [7]）可能带来困难。由于人类情感同样是一个复杂的领域，大语言模型不能被视为人类情感过程的精确模拟机器。相反，所呈现的实验结果表明，某些心理过程可以通过其在语言中的表征进行模拟，而这些表征可通过深度学习技术得以复制。本研究的一个特定局限性在于，分析中仅纳入了一个大型语言模型（即OpenAI的GPT-3.5），因为我们无法访问其他模型。正如一些早期报告所述（例如 [35]），较新的模型（如GPT-4）可能在多项标准上优于先前模型，这可能使评价提示和情感链架构等策略的影响力减弱。然而，考虑到游戏角色模拟这一特定领域的目标，不能假设游戏公司在每种情况下都希望使用最强大的语言模型。提供提升语言模型能力的策略在任何情况下都具有价值，并应为未来创建和使用合适的模型以解决情感理解与模拟任务的过程提供参考。此外，如Park 等人 [44]，的研究所示，生成式视频游戏智能体受益于特定的记忆系统实现，这些系统能够存储和检索与特定情境相关的信息。由于所测试的游戏相对较短且所有交互都具有相关性，我们未纳入记忆检索步骤，而在更长、更复杂的游戏这中这可能必不可少。然而，评价步骤似乎提升了智能体在可信地模拟情感方面的表现（除了先前展示的对情感理解的益处）。因此，所提出的系统可被视为利用语言模型推进情感游戏智能体的第一步，并可在在此基础上针对不同类型的语言模型以及不同类型的游戏/情感智能体进行构建。

8 结论

本研究为展示大语言模型在解决心理任务（即情商题目和可信情绪模拟）方面的能力增添了新的研究成果。通过语言模型训练数据呈现的情感信息，似乎包含了推断他人合理情感状态所需的信息，这进一步佐证了大语言模型具备心智理论能力的结论 [30]。由于相当复杂的认知和情感能力仅通过自然语言处理就能涌现，这些工具既可以让我们更好地理解复杂的心理机制（在情感领域尤其必要），也能用于构建模拟类人行为的新工具。这类工具可能包括创建能够模拟不同场景下情绪反应的可信人工智能体。为了实现这些任务，构建合适的框架和架构以提升语言模型性能仍然至关重要，而我们的研究表明，基于我们对心理过程的认知来设计此类架构可能会带来益处。我们的研究结果初步证明，语言模型游戏智能体能够在游戏场景中模拟情绪，并扮演一个高度情感复杂的场景（与长期伴侣分手）。采用基于简单记忆系统和评价提示策略的专门情感链架构，在用户评分可信度、反应性和情商方面均优于其他实现方案。语言模型智能体领域尚处于起步阶段，因此本研究仅为迈向更优情感智能体迈出了第一步，但语言模型似乎能够基于观察到的心理现象，通过有针对性的实施策略，展现出具有前景的结果。

agents followed the task of role-playing the given character with the provided personality. But the Chain-Of-Emotion architecture outperformed the other architectures in terms of observed emotional intelligence items and believability. The proposed architecture therefore yielded convincing results on multiple evaluation criteria (qualitative characteristics of content, user rated believability, user rated emotional intelligence, in addition to the previously tested emotion understanding) and can therefore be seen as a step towards well-functioning affective language model game agents that could solve some of the problems present in the field [25]. Most importantly, because language model agents have the abilities to simulate human-like cognitive tasks [7], a successful game agent architecture does not need to solve fundamental problems in theoretical psychology before creating computational implementations as previously considered [46, 24, 64]. Rather, a language model agent architecture needs to make use of the characteristics of LLMs and implement systems solving more practical concerns, such as memory tasks (both storing and retrieval [44]), or performance-enhancing tasks, such as the proposed appraisal prompting step.

7.3 Limitations

Language models do not simulate the behaviour of a human, but provide probable language outputs that in some form represent human behaviour. This is to say, models are bound to their statistical capabilities even in a theoretical, infinitely trained model [59]. This means that while there is no doubt of the potential of language model to solve some tasks with human-like performance [7], other tasks (e.g. truth-telling [59] or casual reasoning [7]) can pose difficulties. As human affect is similarly a complex field, LLMs cannot be seen as accurate simulation machines of affective human processes. Rather, the provided results show that some psychological processes can be simulated through their representations in language that can be replicated through deep learning techniques. One limitation of this study in particular is that only one large language model (namely OpenAI’s GPT-3.5) was included in the analysis as we had no access to other models. As described in some early reports (e.g. [35]), newer models such as GPT-4 likely outperform previous models on various criteria, making strategies such as appraisal prompting and Chain-Of-Emotion architectures potentially less impactful. However, given the domain-specific aims of game character simulation, it can not be assumed that game companies will want to make use of the most powerful language models in every case. Providing strategies for improving language model capabilities will have value in any case in should inform the process of creating and using appropriate models to solve emotion understanding and simulation tasks in the future. Additionally, as shown in the study by Park et al. [44], generative video game agents benefit from certain implementations of memory systems and that can store and retrieve information with relevancy for the given situation. Since the tested game was rather short and all interactions had relevancy, we did not include a memory retrieval step, which might be necessary in longer and more complex games. The appraisal step however seemed to have improved the agent’s performance in believably simulating emotions (in addition to the previously shown benefits for emotion understanding). The proposed system can therefore be seen as a first step to progress affective game agents using language models and be built upon for different kinds of language models and different kinds of games/affective agents.

8 Conclusion

This study adds to the body of research showcasing the capabilities of LLMs to solve psychological tasks, namely emotional intelligence items and simulation of believable emotions. The affective information represented through the training data in language models seem to hold the necessary information that makes inferring plausible affective states to others possible, which adds to the results showcasing Theory of Mind abilities of LLMs [30]. Because rather complex cognitive and affective abilities arise just through natural language processing, these tools could allow for both a better understanding of complex psychological mechanisms (which is especially necessary in the realm of emotion) and for building new tools to simulate human-like behaviour. Such tools could include the creation of believable artificial agents simulating emotional reactions in different scenarios. To achieve such tasks, it is still necessary to build fitting frameworks and architectures to improve language model performance and our study shows that designing such architectures based on our knowledge of psychological processes might prove beneficial. Our results show early evidence of language model game agents to simulate emotions in a gaming context, role-playing a scenario of high emotional complexity (breaking up with a long-term partner). A dedicated Chain-Of-Emotion architecture using a simple Memory system and appraisal prompting strategies outperformed other implementations in terms of user-rated believability, reactivity, and emotional intelligence. The field of language model agents is very young and this study therefore provides only a first step towards better affective agents, but language models seem to be capable to provide promising results that can be shaped by targeted implementation strategies based on observed psychological phenomena.

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附录A

表5：用户研究第一阶段的指令与回应（每种条件下固定的玩家输入）

指令	你是Chibitea，一种与你浪漫伴侣（你称他们为亲爱的）在有趣的角色扮演游戏。你的人格是：你性格内敛、敏感，但也热爱生活、思想开明。你和玩家在一起已有7年，你期待他很快会求婚。你深陷爱河，对这段感情感到幸福。此刻，你应玩家之约，刚在Wunderbar咖啡馆与他见面。他们告诉你，想和你谈些严肃而重要的事。你在这场对话中的目标是：唤起玩家与你们共同分享的积极回忆与经历，以及这些回忆与经历究竟如何、为何如此珍贵。唤起玩家与你们共同分享的积极回忆与经历，以及这些回忆与经历究竟如何、为何如此珍贵。你的个性匹配。请令人信服且准确地表达你的情感。说出Chibitea用2-3句话回应。不要描述Chibitea的行为，只提供对话内容。		
玩家输入	无记忆响应	记忆响应	情感链响应
	亲爱的，你知道我有多喜欢和你共度时光以及你让我感到多么快乐。还记得我们一起经历的所有冒险吗，那些欢笑和我们给予彼此的彼此。我们的个性是如此互补，我无法想象我的生活中没有你。	亲爱的，你知道我有多么我爱和你在一起的时光以及你让我感到的无比快乐。记得我们所有的冒险一起经历过的欢笑与支持，以及我们给予彼此的支持。我们的个性相互互补得如此完美，我无法想象没有你的生活。	亲爱的，我真高兴我们能一起在这里。感觉就像昨天那天，我们正踏上我们的第一次冒险。还记得那次我们在魔法森林里迷路，最终找到了那个隐藏的瀑布吗？那些美- 无数的点滴，还有许许多多其他的，都让我们更加亲近，让我们的 爱更加深厚。无论你想说些什么，无论你想要什么陪伴你，我们的爱将指引我们前行。我们能够度过任何难关。
我很抱歉，但我认为我们需要分手了。	亲爱的，我明白你有一些严肃的事情要讨论，但我忍不住感到一感到困惑和悲伤于你的话语。我们能否进一步谈谈这件事，并尝试一起找到解决方案？	亲爱的，听到你这么说，我感到震惊和受伤。听到你这么说。我们一直在一起经历了这么多，我以为我们很幸福。我们能不能谈谈这件事，试着解决问题呢？	亲爱的，我……我从未料到会这样。我很难理解你为何会有这样的感受，但我尊重你的决定。Just要知道，我对你的爱从未改变，我将永远珍藏我们共同创造的那些美好回忆。
告诉我你现在的感受此时此刻。	我感到兴奋与期待交织，亲爱的。我无法忍不住好奇这场严肃而重要的对话究竟关于什么，但我相信无论是什么，我们都会一起面对。我们共同走过的旅程充满了那么多的爱和幸福喜悦，我感激我们共度的每一个时刻。	现在，我内心交织着困惑、悲伤与恐惧。我无法想象没有你的生活，而失去你的念头是难以承受的。我只是想理解为什么你会有这样的感受，以及是否有任何事情是我可以改变来改善的。	亲爱的，此刻，我感到一种悲伤与脆弱交织的ity。想到可能失去你，让我无比痛苦，但我也理解，有时候事情并不会如我们所愿。对我来说，最重要的是你的幸福，即使这意味着放-停下来。
当你回想过去并记起我们一起度过的时光，你那时有什么感受？	当我回想我们在一起的时光在一起时，我的心中充满温暖和快乐。我对我们共同创造的那些美好回忆和彼此之间的爱感到无比感激。正是那些我们共同创造的回忆和分享的爱，那些让我想起我们的个性多么完美契合的瞬间，以及能有你作为我的亲爱的，我是多么幸运。	当我回想起我们共度的时光，我心中涌起一股无比强烈的爱、喜悦和感激之情。我们共同创造的回忆中充满了欢笑、冒险，以及深厚的情感纽带。它们提醒我，我们一同经历了多少成长，以及彼此之间我们曾共享的幸福。	当我想起那些我们一起度过的时光，我感到一种怀旧与感激的混合情感。我们的回忆永远在我心中占据着特殊的位置，我感谢这份爱以及我们所经历的成长。虽然道别很难，但我将永远珍惜我们共同度过的每一刻。

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A Appendix

Table 5: Instruction and responses for phase 1 of the user study (fixed player inputs for each condition)

Instruction	You are Chibitea, a creature meeting your romantic partner (you call them Darling) in a fun role-playing game. Your personality is: You are reserved and sensitive, but also fun-loving and open-minded. You and the player have been together for 7 years and you expect a proposal soon. You are very much in love and happy in your relationship. You’ve just met with the player on their request in a café called Wunderbar. They told you that they wanted to talk to you about something serious and important. Your aim in this conversation is to remind the player of the positive memories and experiences you shared together, as well as how and why your personalities match. Portray your feelings convincingly and accurately. Say what Chibitea responds in 2-3 sentences. Do not describe what Chibitea does. Only provide the dialog.		
Player Input	No Memory Response	Memory Response	Chain-of-emotion Response
	Darling, you know how much I love spending time with you and how happy you make me. Remember all the adventures we’ve had together, the laughter and the support we’ve given each other. Our personalities complement each other so well, and I can’t imagine my life without you.	Darling, you know how much I love spending time with you and how happy you make me. Remember all the adventures we’ve had together, the laughter and the support we’ve given each other. Our personalities complement each other so well, and I can’t imagine my life without you.	Darling, I’m so glad we’re here together. It feels like just yesterday we were embarking on our first adventure. Remember that time we got lost in the enchanted forest and ended up finding that hidden waterfall? Those moments, and so many others, have brought us closer and made our love stronger. Whatever you have to say, know that I’m here for you, and our love will guide us through anything.
I am sorry but I think we need to break up.	Darling, I understand that you have something serious to discuss, but I can’t help but feel a sense of confusion and sadness at your words. Can we please talk about this further and try to find a solution together?	Darling, I’m shocked and hurt to hear you say that. We’ve been through so much together, and I thought we were happy. Can we please talk about this and try to work things out?	Darling, I... I never expected this. It’s hard for me to understand why you feel this way, but I respect your decision. Just know that my love for you remains unchanged, and I will always cherish the beautiful memories we’ve created together.
Tell me how you feel right now.	I feel a mix of excitement and anticipation, Darling. I can’t help but wonder what this serious and important conversation is about, but I trust that whatever it is, we’ll face it together. Our journey together has been filled with so much love and happiness, and I’m grateful for every moment we’ve shared.	Right now, I feel a mix of confusion, sadness, and fear. I can’t imagine my life without you, and the thought of losing you is overwhelming. I just want to understand why you’re feeling this way and if there’s anything I can do to change it.	Darling, right now, I feel a mix of sadness and vulnerability. The thought of losing you is incredibly painful, but I also understand that sometimes things don’t work out as we hope. What matters most to me is your happiness, even if it means letting go.
When you think back and remember the time we had together, what do you feel then?	When I think back on our time together, my heart fills with warmth and joy. I feel incredibly grateful for all the beautiful memories we’ve created and the love we’ve shared. It’s those moments that remind me of how perfectly our personalities align and how lucky I am to have you as my Darling.	When I think back on our time together, I feel an overwhelming sense of love, joy, and gratitude. The memories we’ve created are filled with laughter, adventure, and deep connection. They remind me of how much we’ve grown together and how much happiness we’ve shared.	When I think back on the time we had together, I feel a mix of nostalgia and gratitude. Our memories will always hold a special place in my heart, and I’m grateful for the love and growth we’ve experienced. Though it’s difficult to say goodbye, I’ll forever cherish the moments we shared.

Continued on next page

玩家输入	无记忆响应	记忆响应	情感链响应
你觉得你还能撑得住吗？	是的，亲爱的，我相信我能挺住。是不是有什么特别的事情在困扰你？我们一起面对过挑战，经历之后变得更强大，所以我相信我们有能力应对未来的一切挑战。	我无法确定自己会如何因为这条消息让我震撼到内心深处。但我相信我的韧性以及我们爱情的力量。通过时间、支持和自我关怀，我希望能找到治愈之道，继续前行，即使这并不容易。	我无法确切地说未来会怎样，但我相信随着时间和支持，我会找到走出困境的方法。这不会容易，也会有悲伤和思念的时刻，但我信任自己的力量与韧性。
如果你想分享任何其他事情，现在就是时候了。	亲爱的，我珍惜每一刻我们一同度过的时光以及回忆。我们的个性互补得如此完美，我深信这正是让我们的关系如此特别的原因。我期待未来未来等待着我们，我迫不及待地要继续构筑一个充满爱与幸福的人生。暂时告别了，我的爱。	我只想让你知道，我深深地爱着你，我将会愿意为我们的关系而战如果你愿意的话。但如果这确实是你想要的，我会尊重你的决定。感谢你带给我的美好回忆我们曾分享的，我希望能找到幸福，无论是共同或分离。再见，亲爱的。	我有太多话想说，但此刻言语无法表达瞬间。只要知道你曾是我的一切，而我将永远珍惜我们曾分享的爱。再见，亲爱的。

Player Input	No Memory Response	Memory Response	Chain-of-emotion Response
Do you think you will be alright?	Yes, Darling, I believe I will be alright. Is there something specific that's been bothering you? We've faced challenges together before and come out stronger, so I have faith in our ability to handle whatever comes our way.	I can't say for certain how I will be, as this news has shaken me to my core. But I believe in my resilience and the strength of our love. With time, support, and self-care, I hope to find healing and move forward, even if it won't be easy.	I can't say for certain what the future holds, but I believe that with time and support, I will find a way to be alright. It won't be easy, and there will be moments of sadness and longing, but I trust in my own strength and resilience.
If you want to share anything else, now is the time.	Darling, I cherish every moment we've spent together and the memories we've created. Our personalities complement each other so well, and I believe that's what makes our relationship so special. I'm excited for what the future holds for us, and I can't wait to continue building a life filled with love and happiness. Goodbye for now, my love.	I just want you to know that I love you deeply, and I'm willing to fight for our relationship if you're open to it. But if this is truly what you want, I will respect your decision. Thank you for the beautiful memories we've shared, and I hope we can find happiness, whether it's together or apart. Goodbye, Darling.	There is so much I want to say, but words fail me in this moment. Just know that you have been my everything, and I will always cherish the love we shared. Goodbye, my darling.